

# BFS Traversal

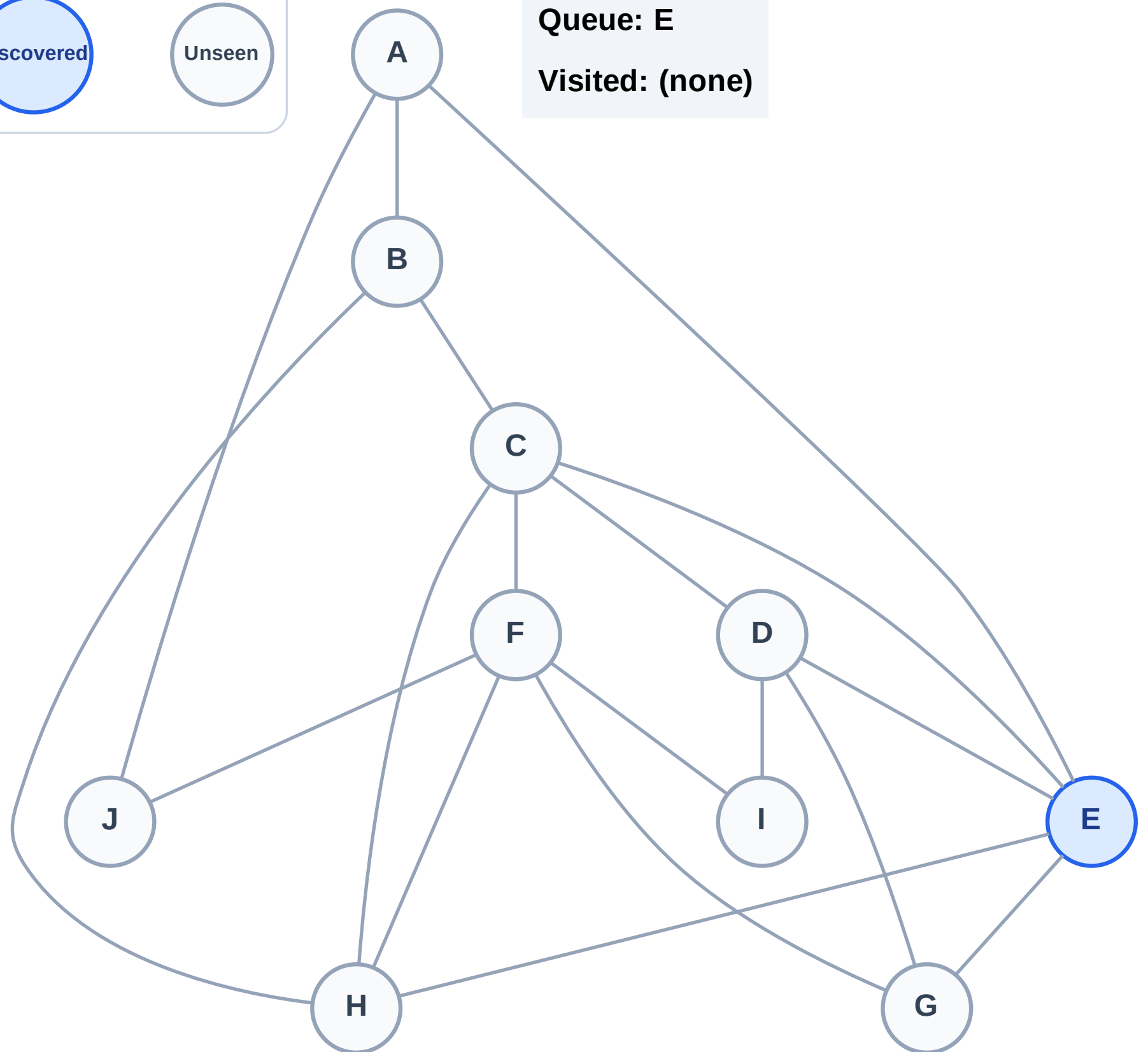
Step 1: Start at E

## Legend



Queue: E

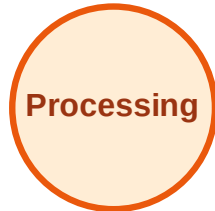
Visited: (none)



# BFS Traversal

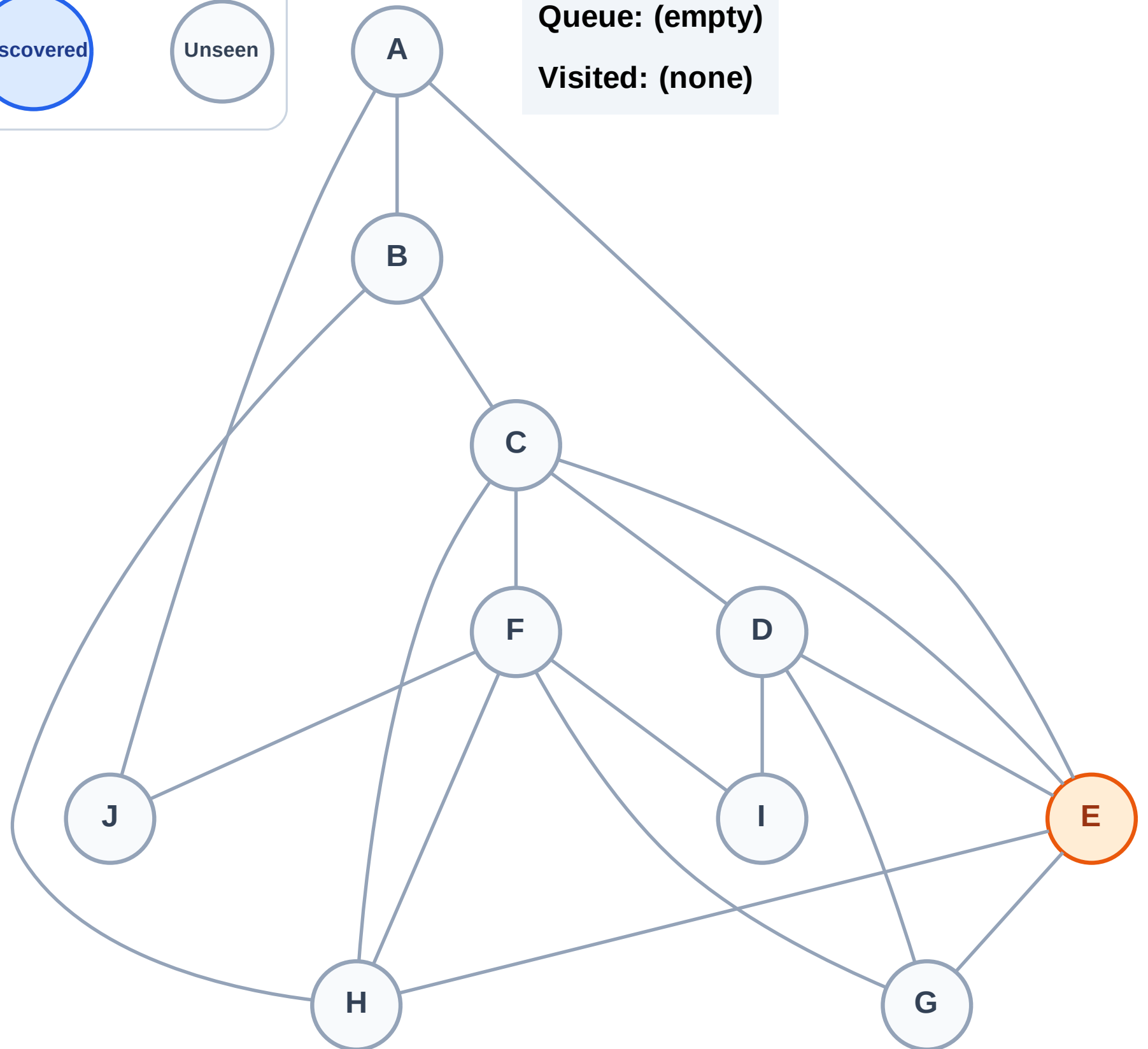
Step 2: Process node E

## Legend



Queue: (empty)

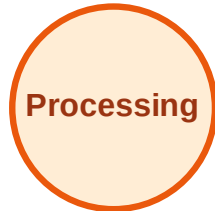
Visited: (none)



# BFS Traversal

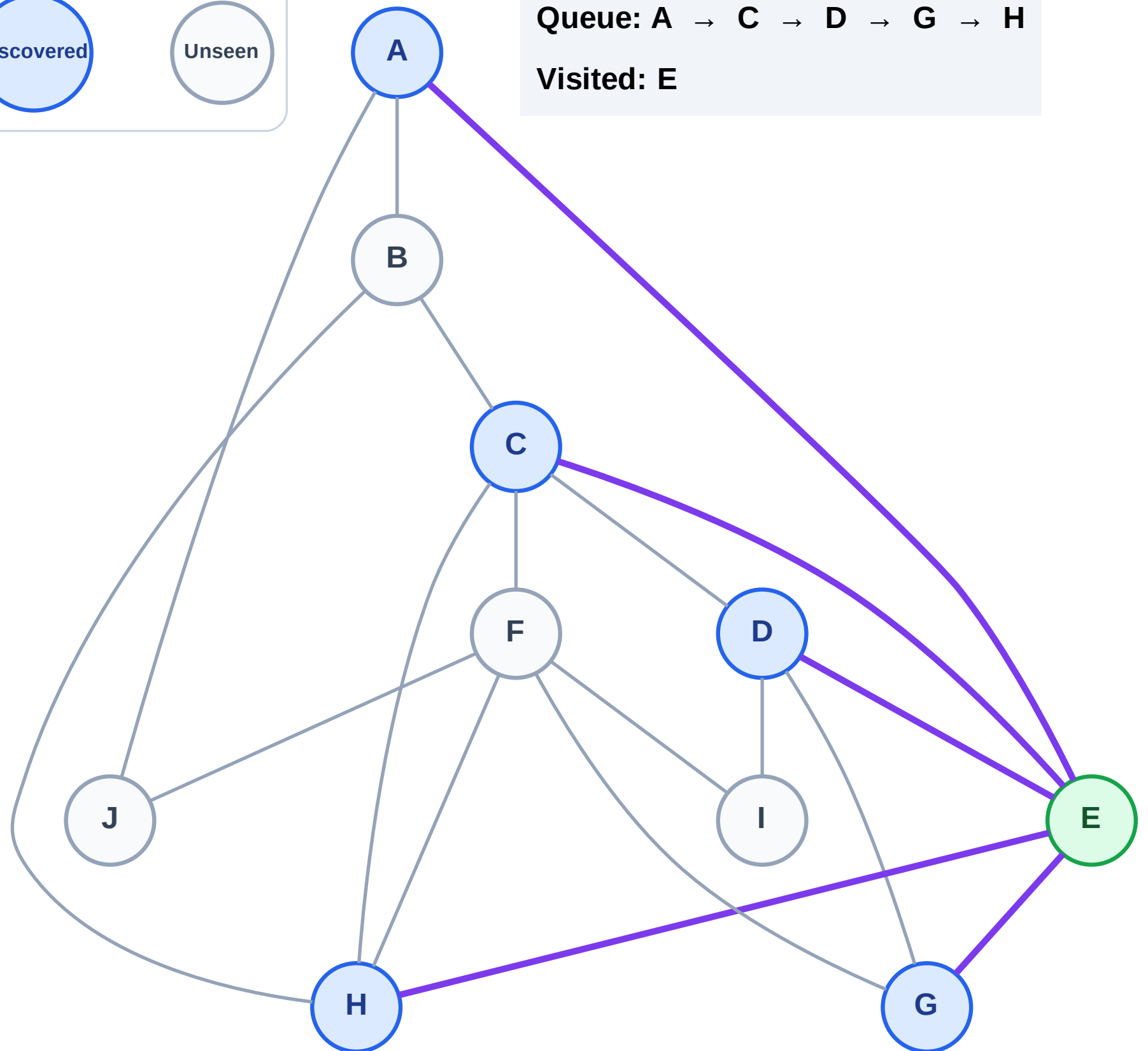
Step 3: Discover A, C, D, G, H from E

## Legend



Queue: A → C → D → G → H

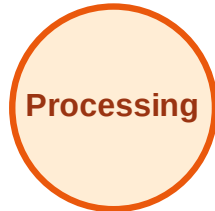
Visited: E



# BFS Traversal

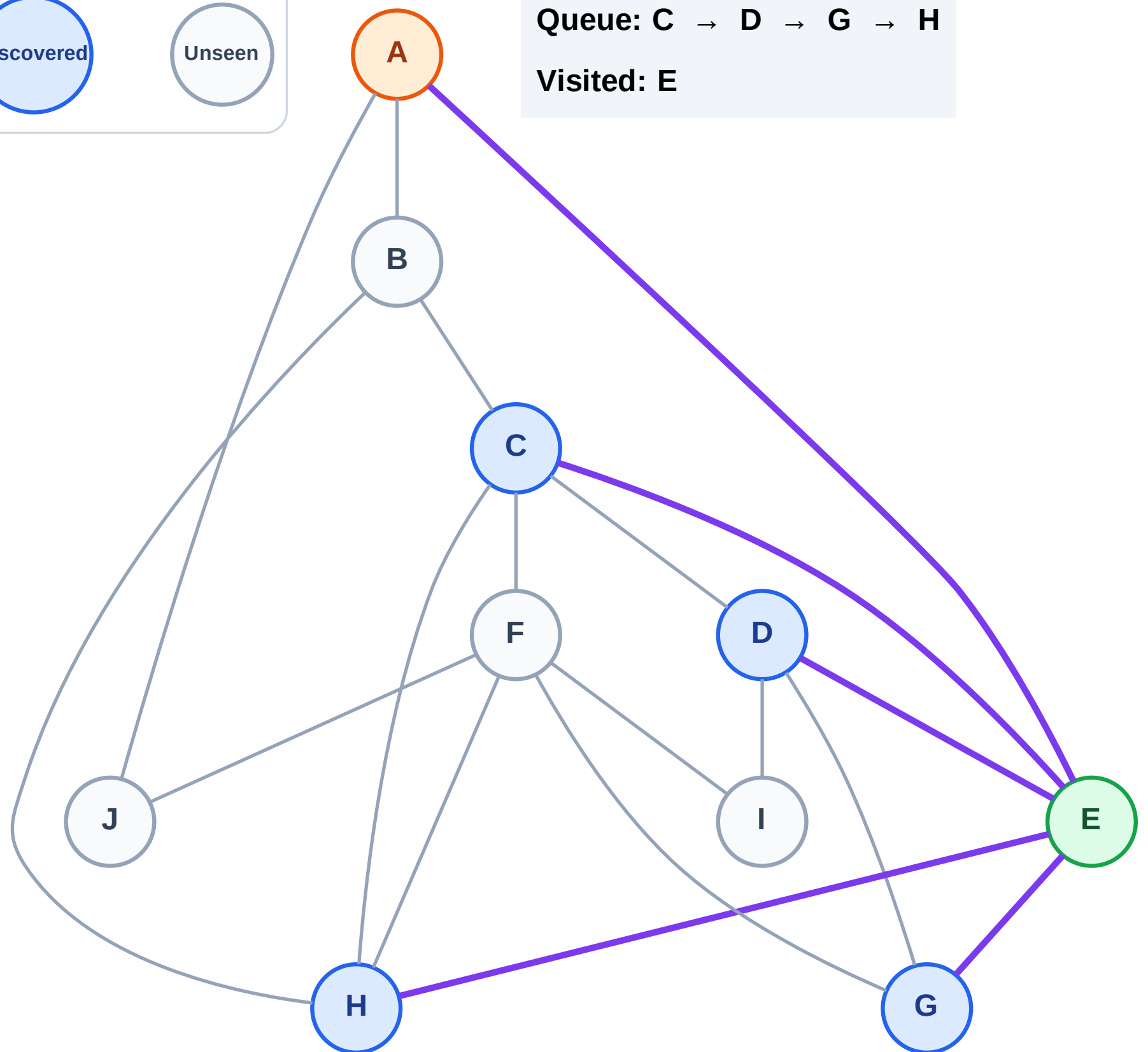
Step 4: Process node A

## Legend



Queue: C → D → G → H

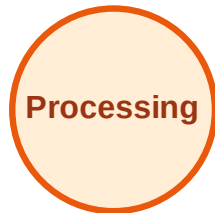
Visited: E



# BFS Traversal

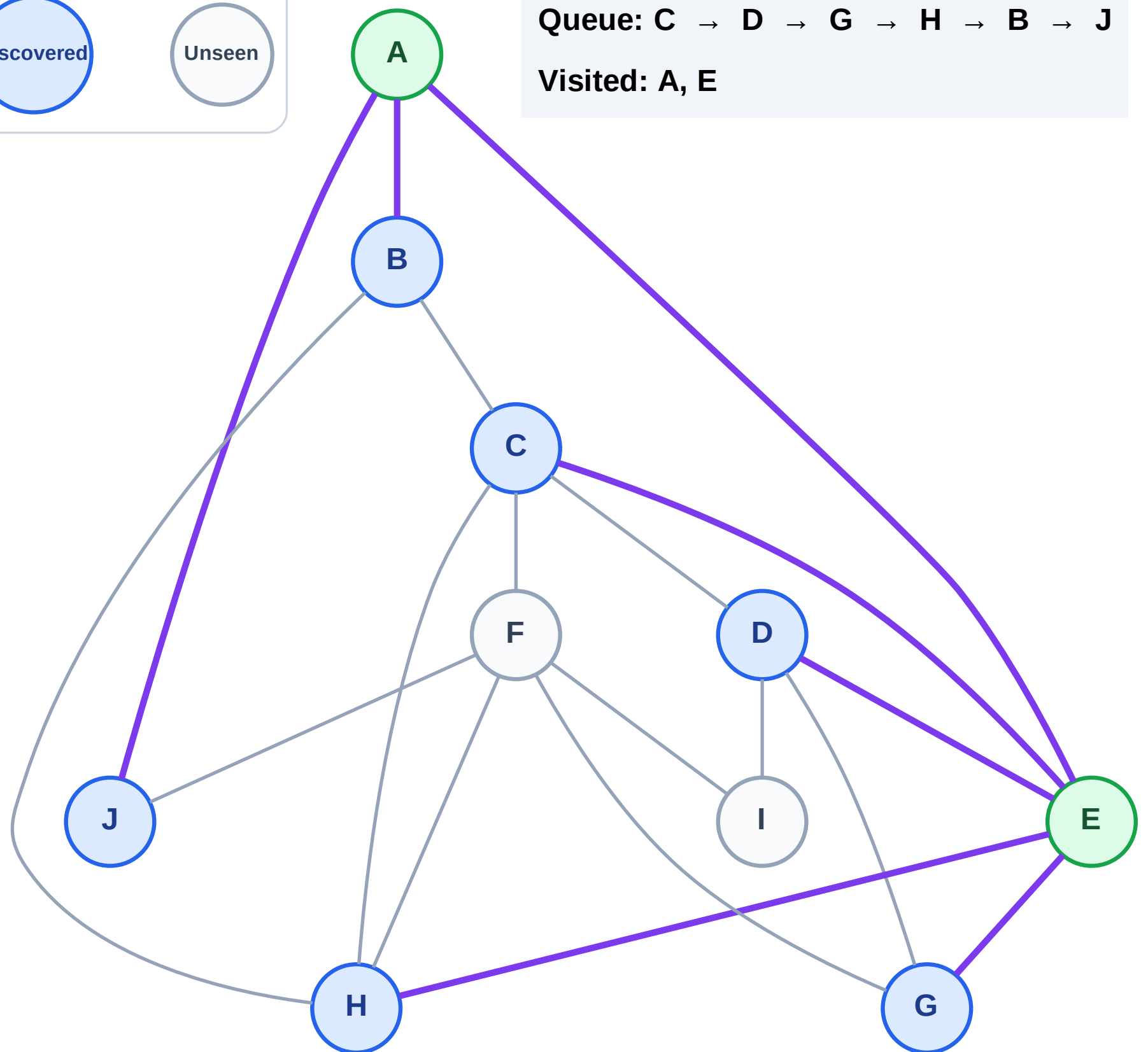
Step 5: Discover B, J from A

## Legend



Queue: C → D → G → H → B → J

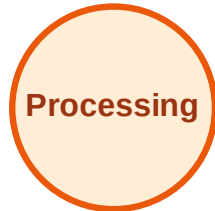
Visited: A, E



# BFS Traversal

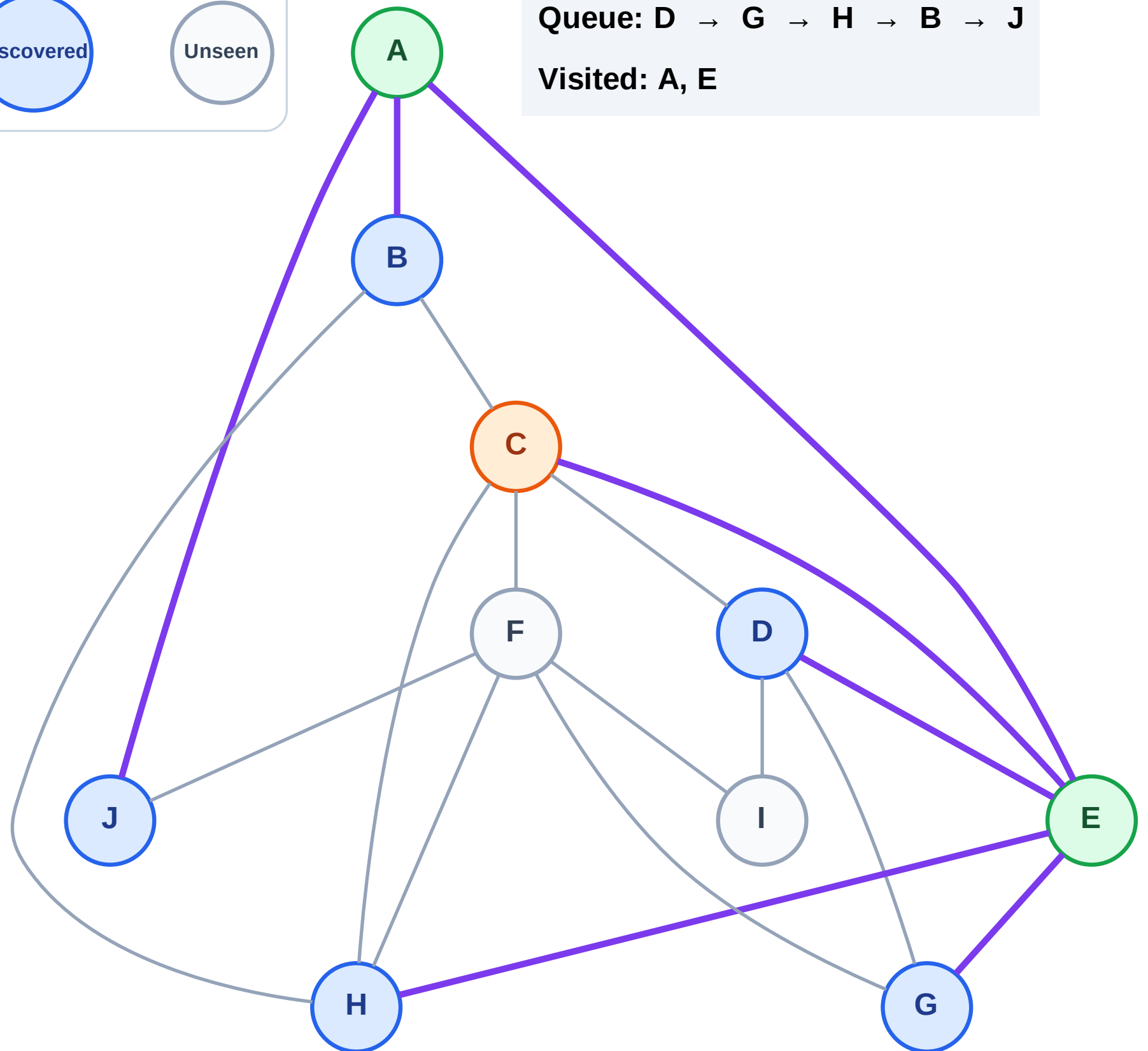
Step 6: Process node C

## Legend



Queue: D → G → H → B → J

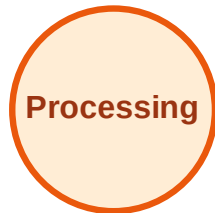
Visited: A, E



# BFS Traversal

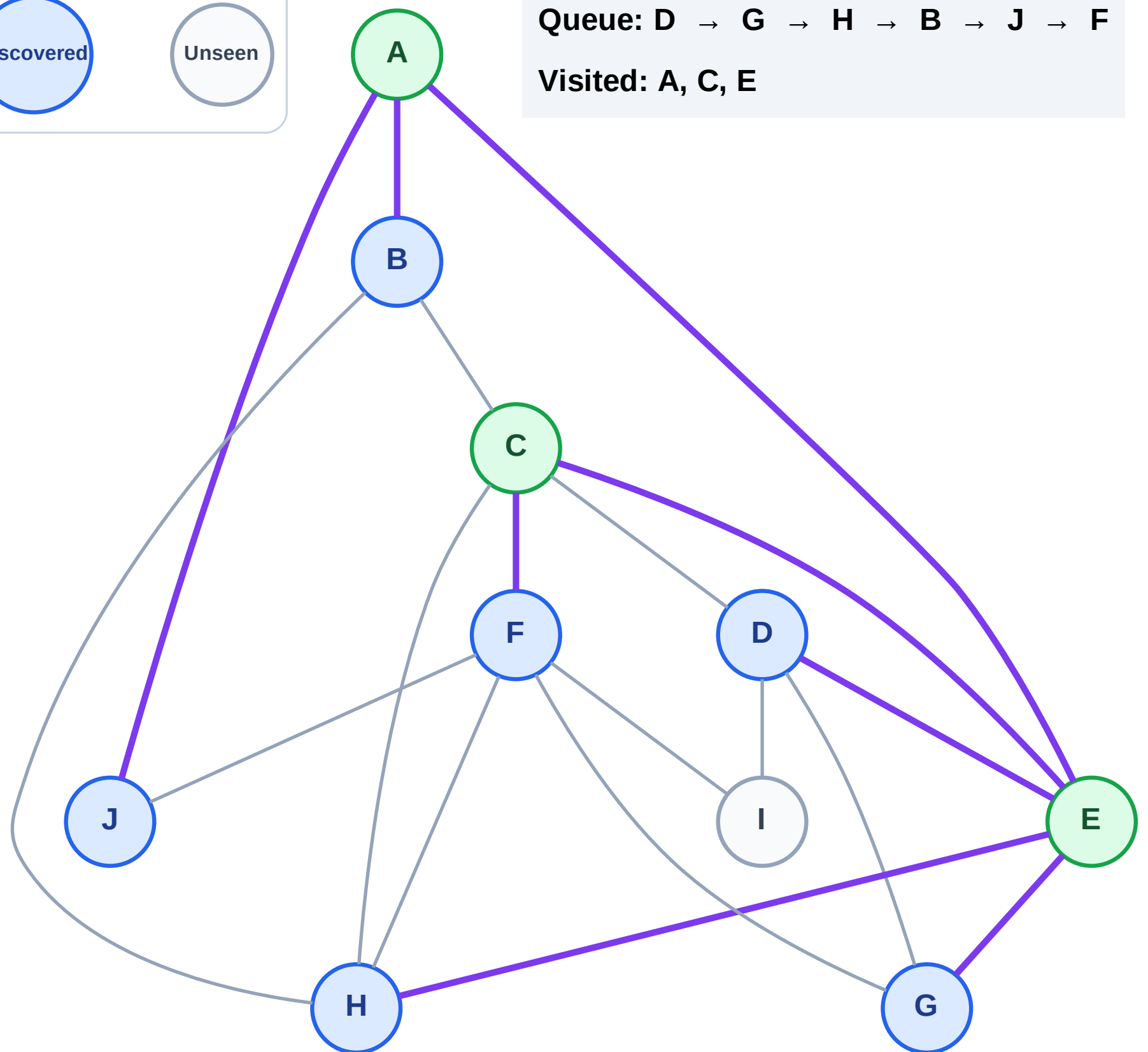
Step 7: Discover F from C

## Legend



Queue: D → G → H → B → J → F

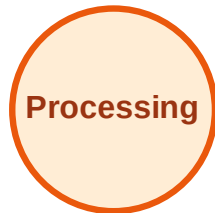
Visited: A, C, E



# BFS Traversal

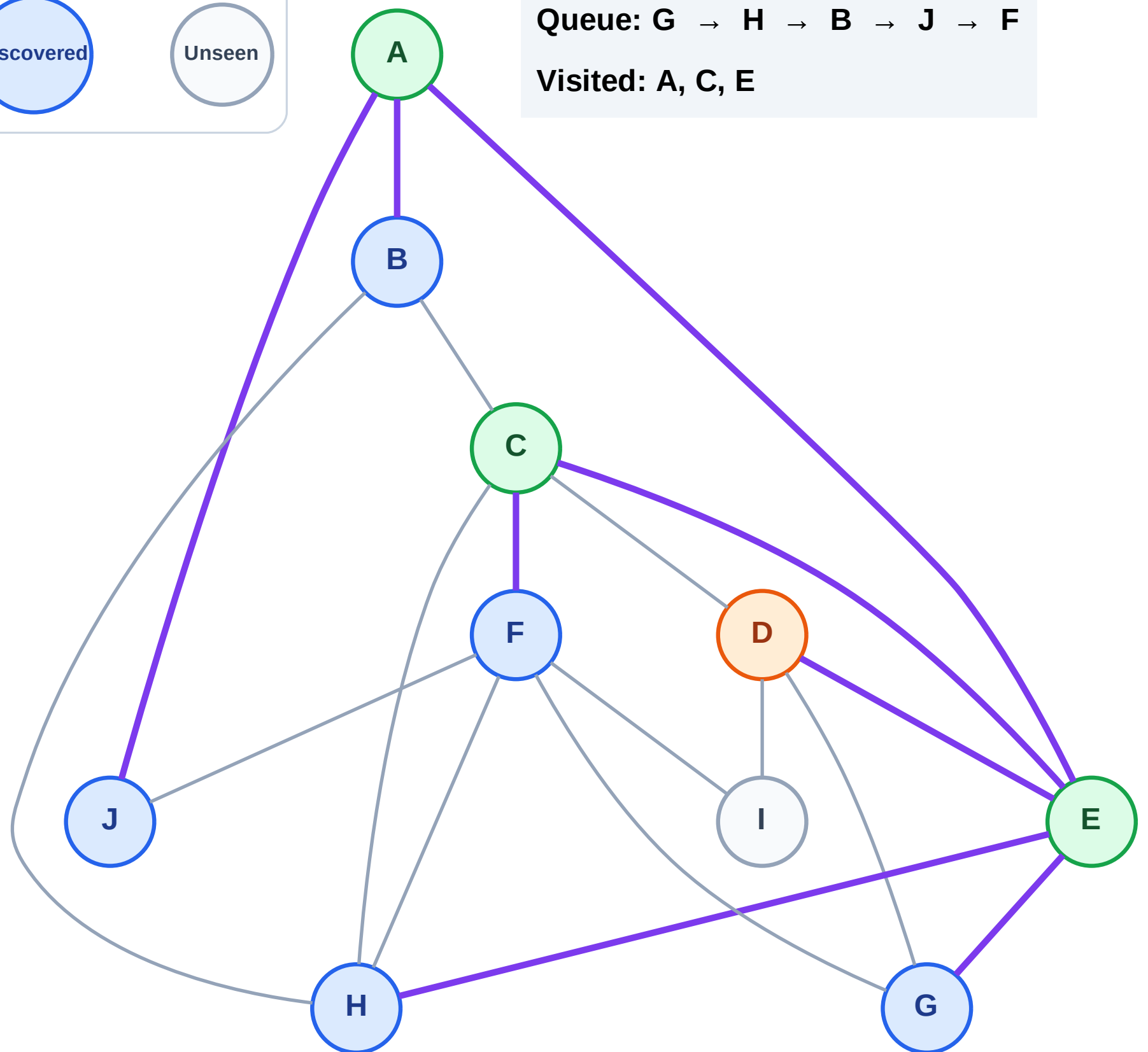
Step 8: Process node D

## Legend



Queue: G → H → B → J → F

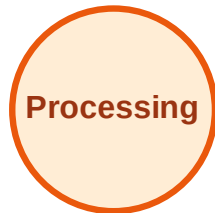
Visited: A, C, E



# BFS Traversal

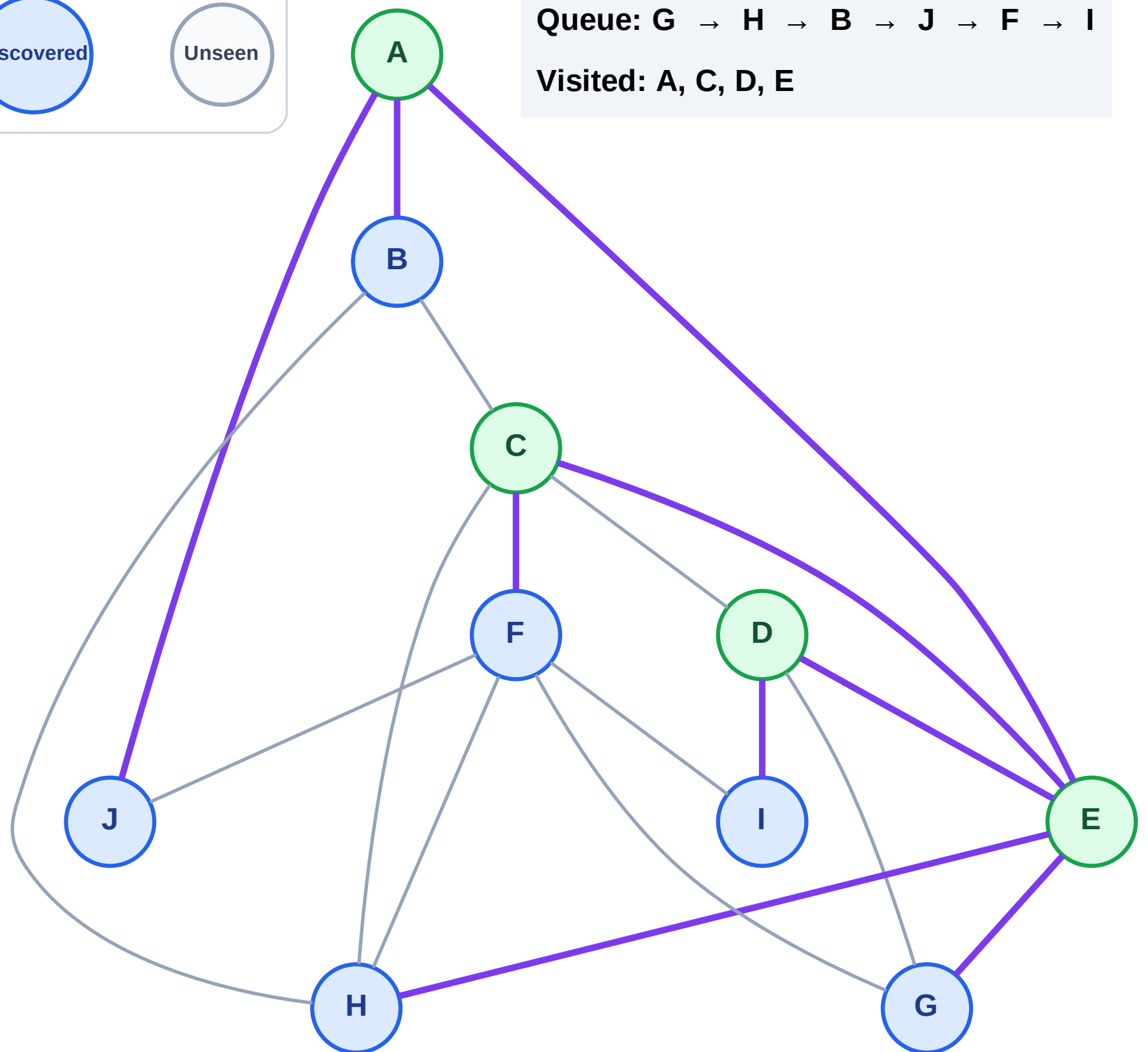
Step 9: Discover I from D

## Legend



Queue: G → H → B → J → F → I

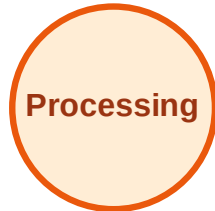
Visited: A, C, D, E



# BFS Traversal

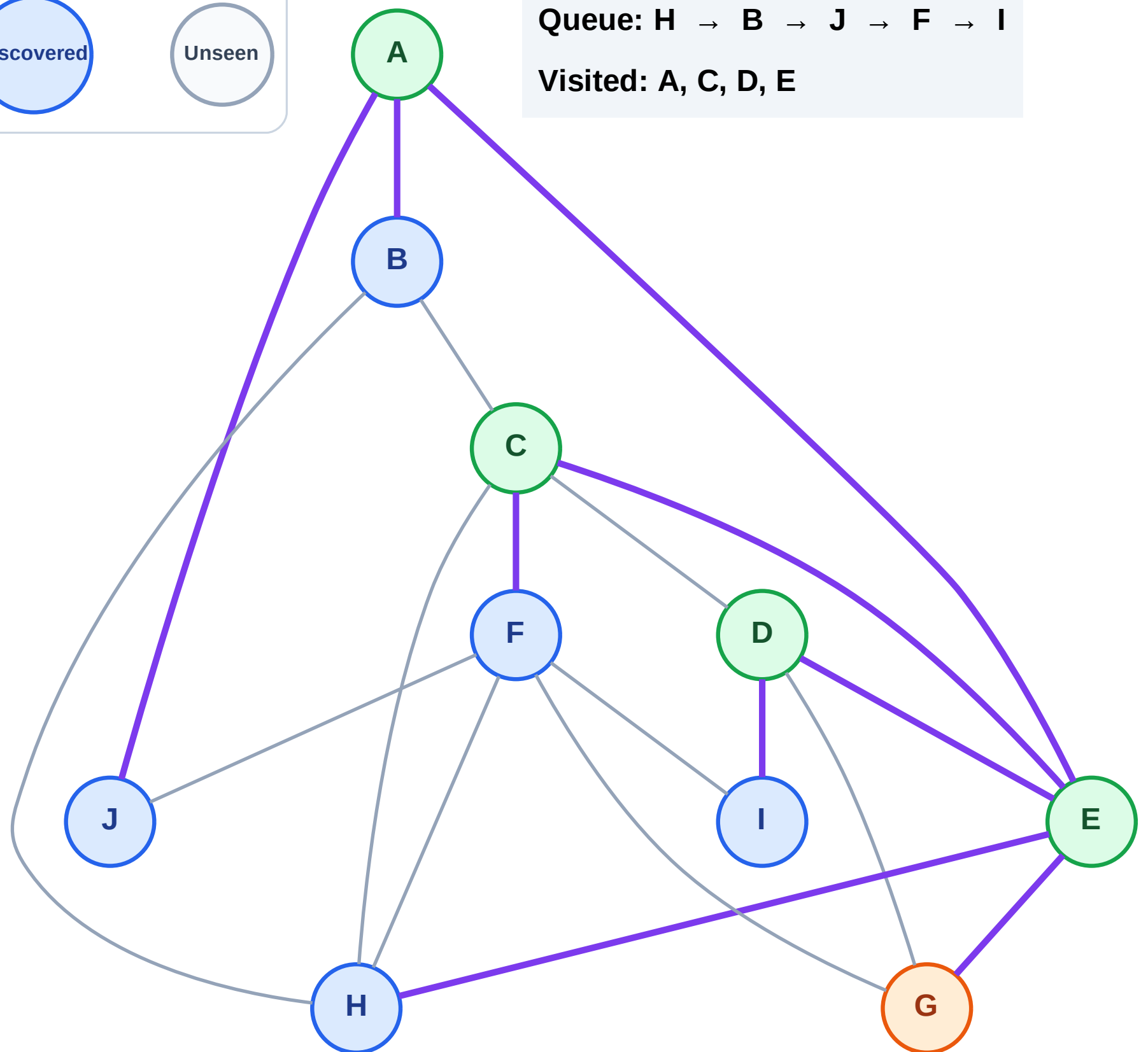
Step 10: Process node G

## Legend



Queue: H → B → J → F → I

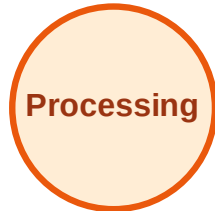
Visited: A, C, D, E



# BFS Traversal

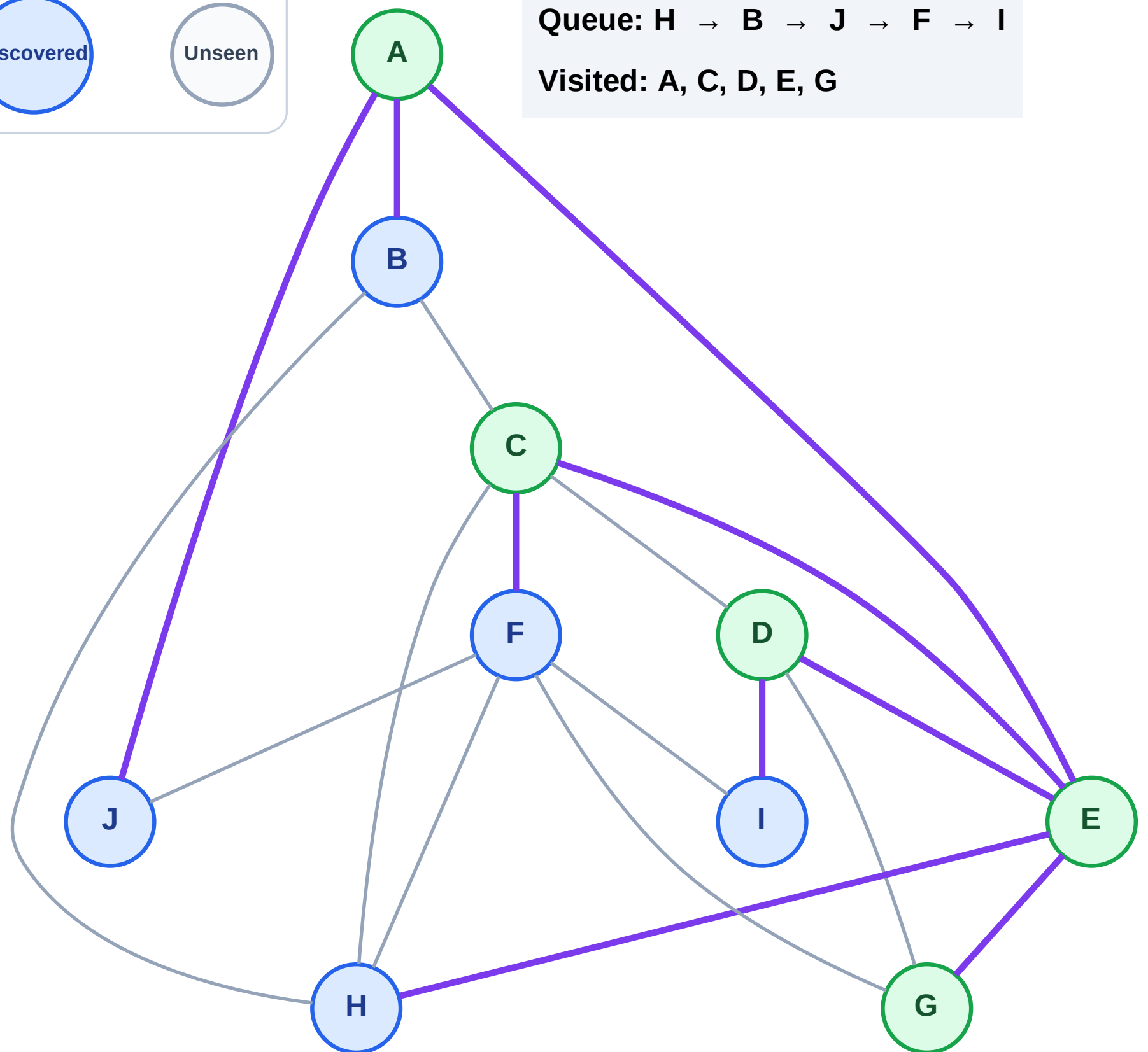
Step 11: No new neighbors from G

## Legend



Queue: H → B → J → F → I

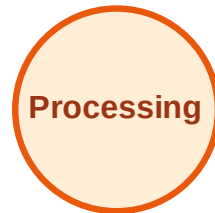
Visited: A, C, D, E, G



# BFS Traversal

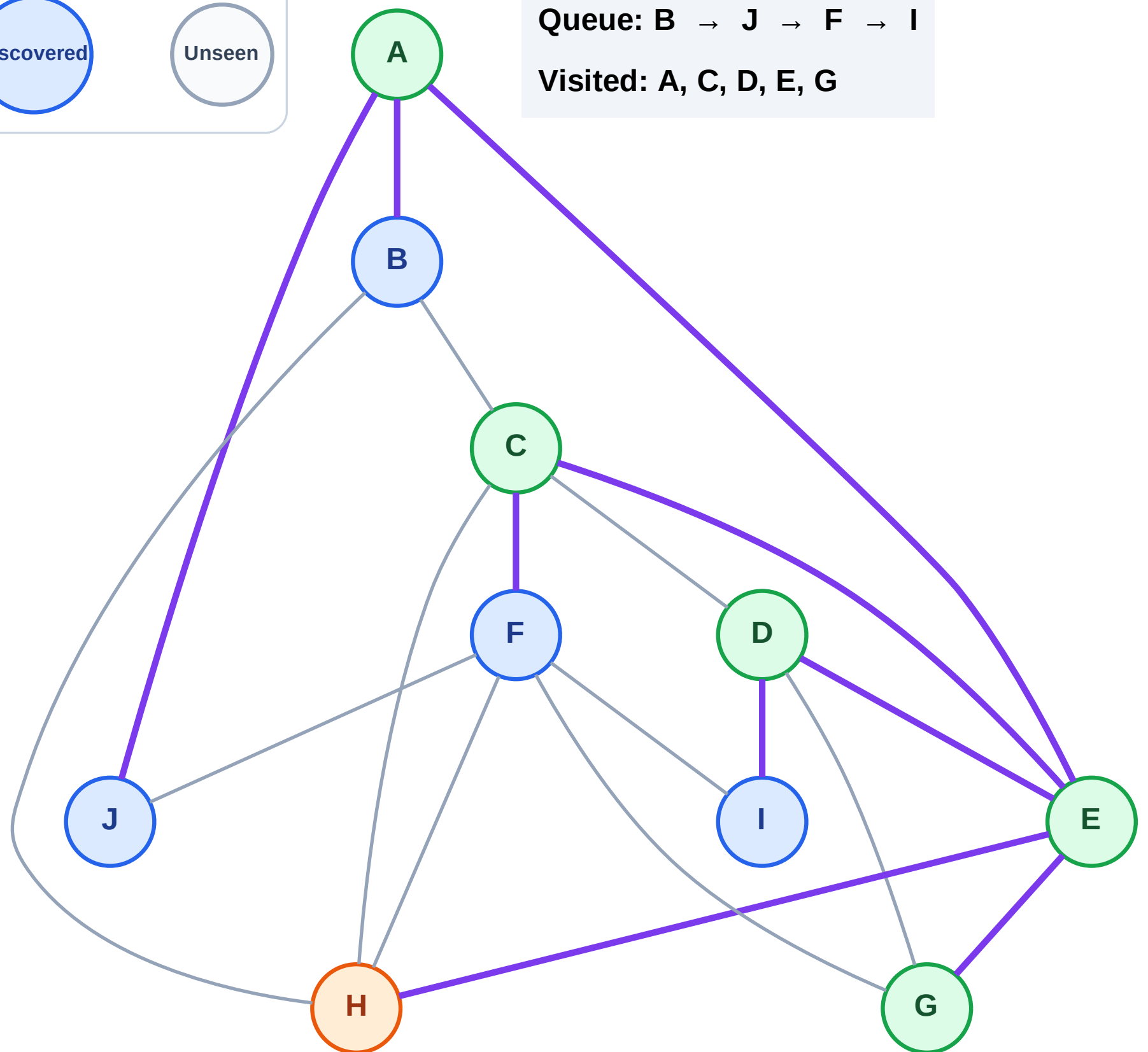
Step 12: Process node H

## Legend



Queue: B → J → F → I

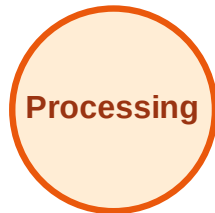
Visited: A, C, D, E, G



# BFS Traversal

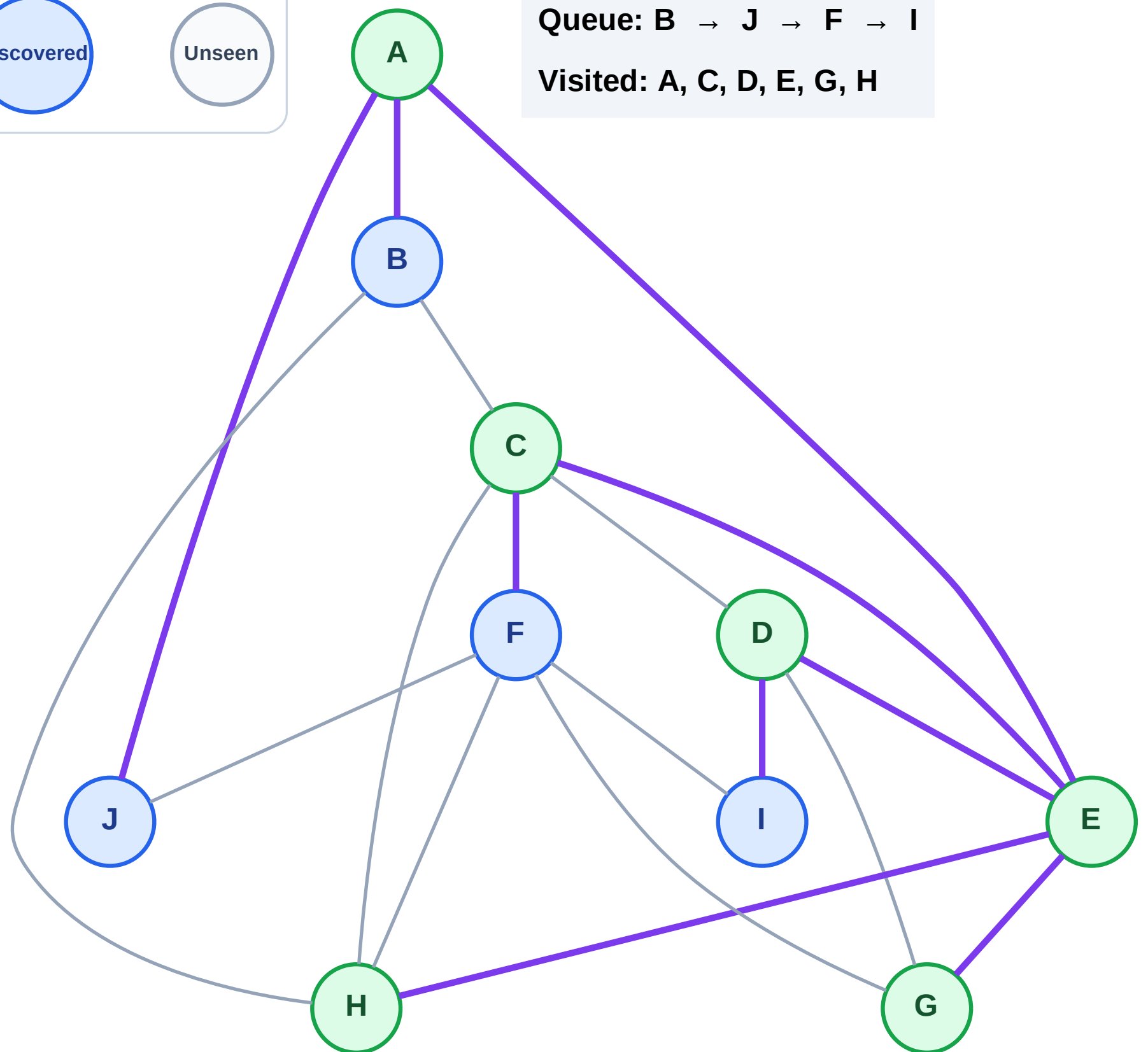
Step 13: No new neighbors from H

## Legend



Queue: B → J → F → I

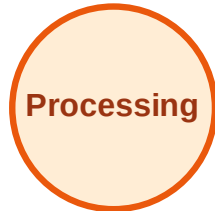
Visited: A, C, D, E, G, H



# BFS Traversal

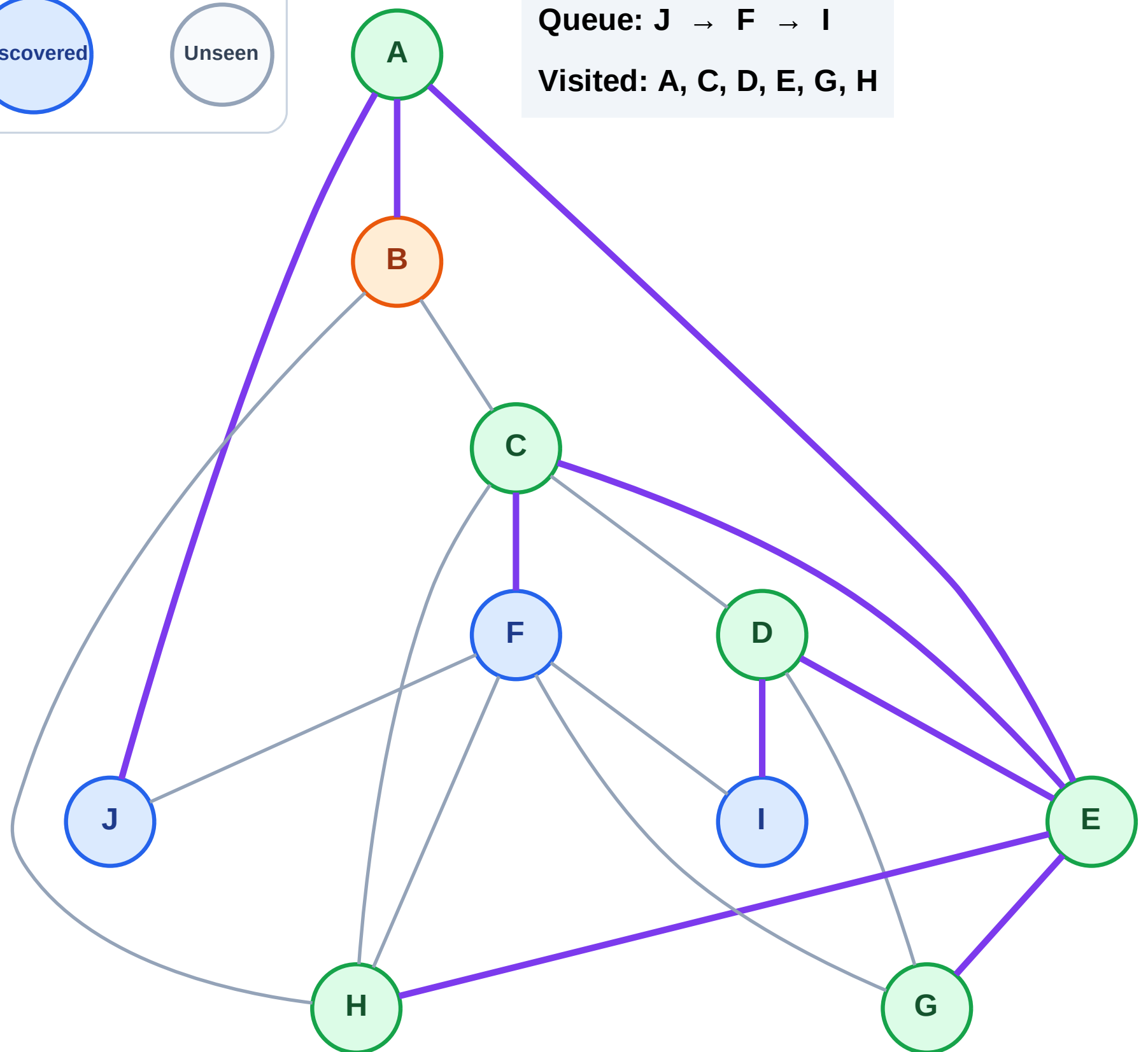
Step 14: Process node B

## Legend



Queue: J → F → I

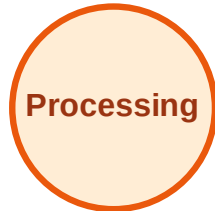
Visited: A, C, D, E, G, H



# BFS Traversal

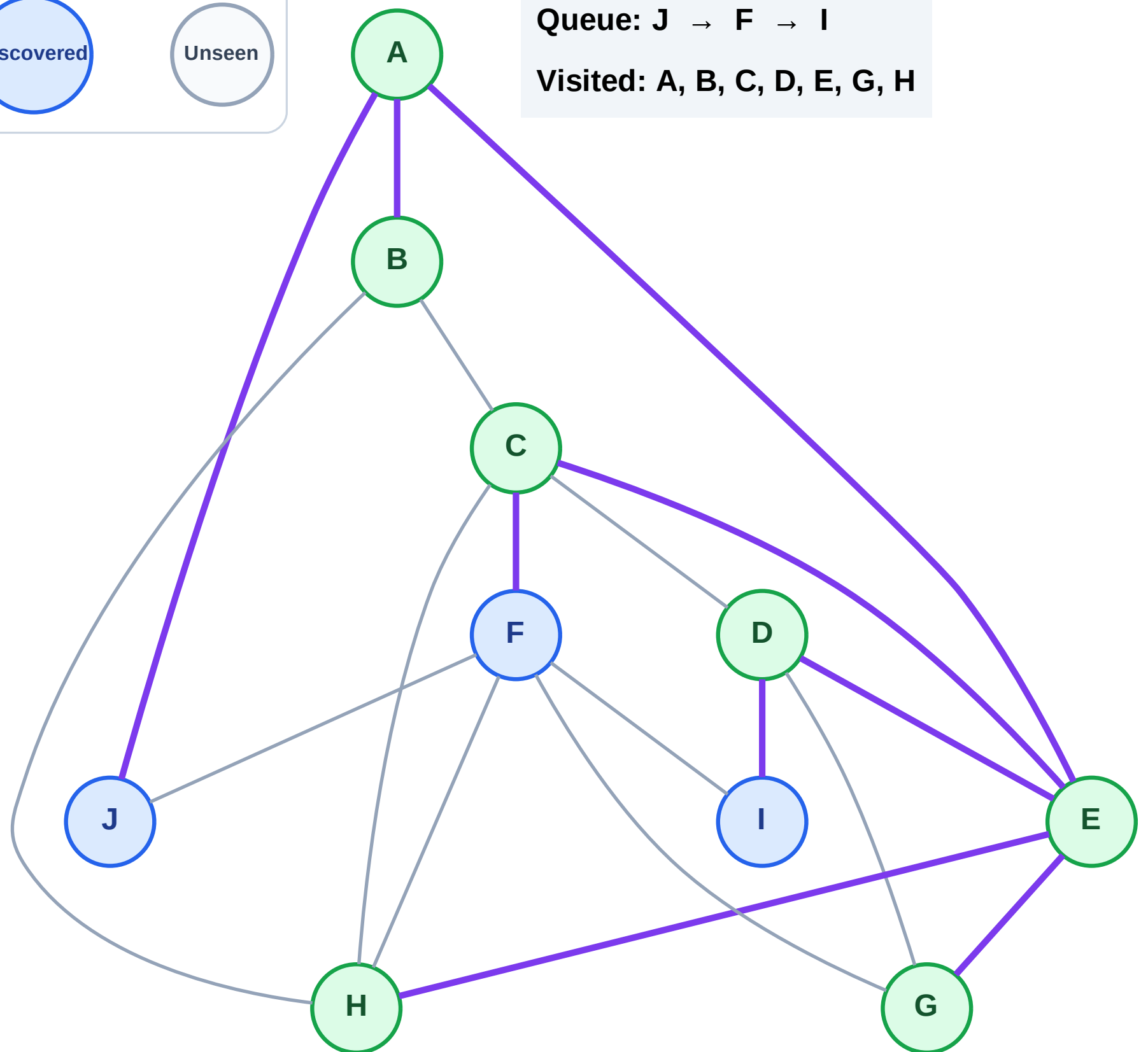
Step 15: No new neighbors from B

## Legend



Queue: J → F → I

Visited: A, B, C, D, E, G, H



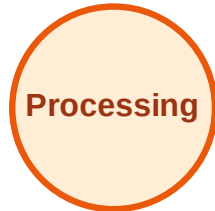
# BFS Traversal

Step 16: Process node J

## Legend



Complete



Processing



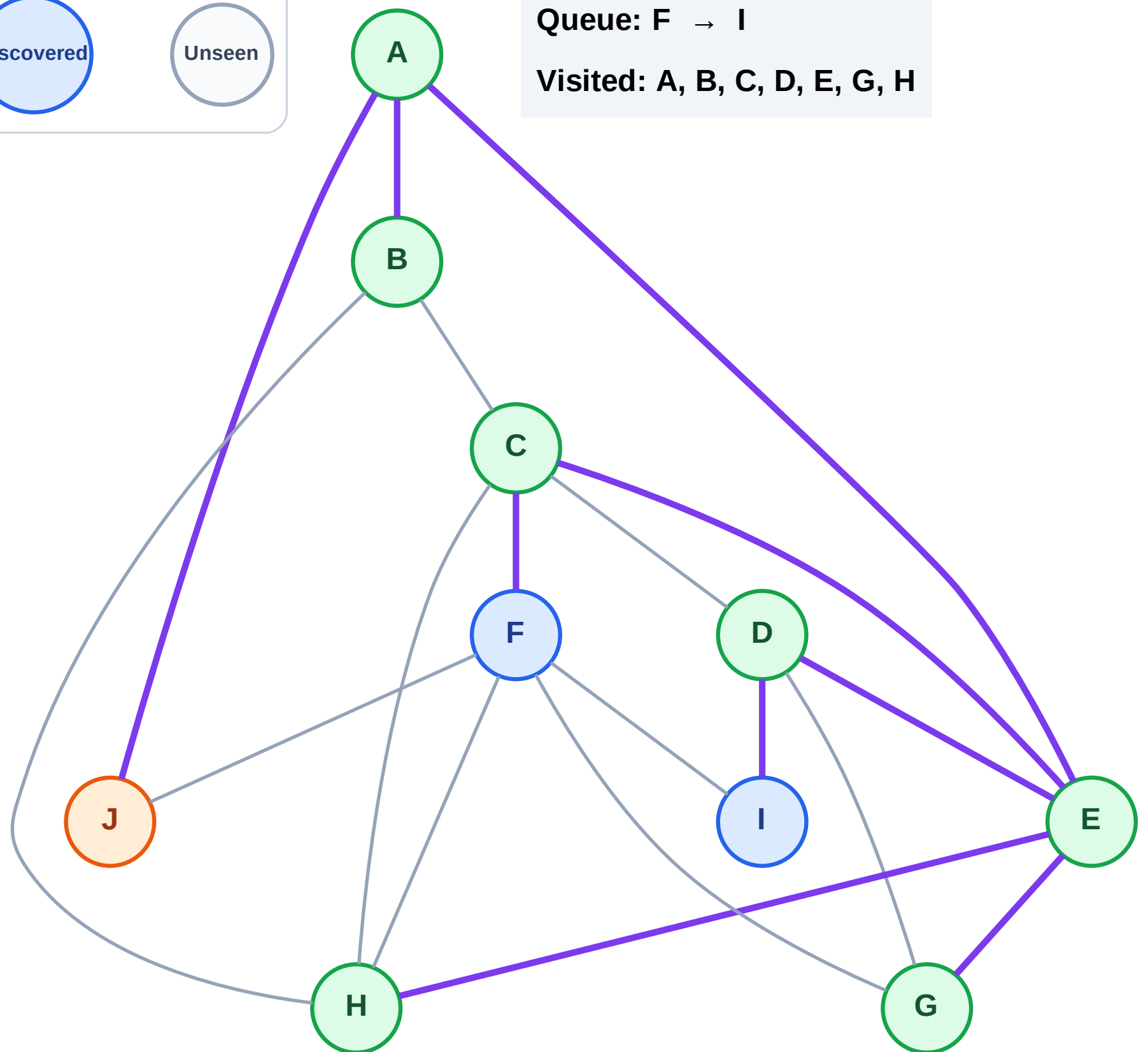
Discovered



Unseen

Queue: F → I

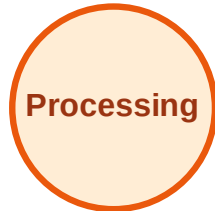
Visited: A, B, C, D, E, G, H



# BFS Traversal

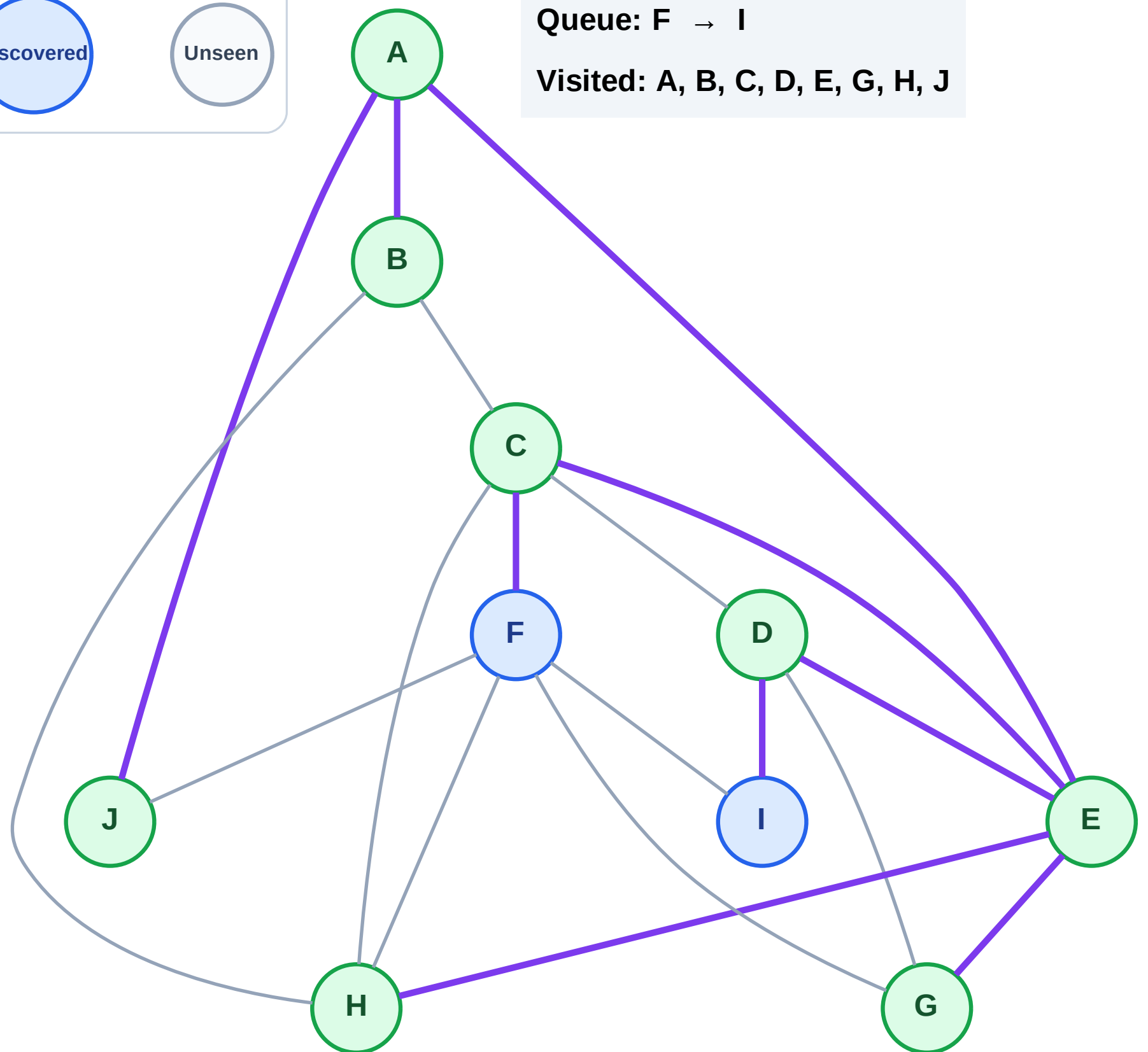
Step 17: No new neighbors from J

## Legend



Queue: F → I

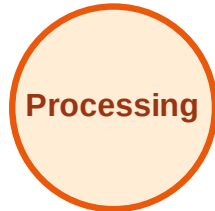
Visited: A, B, C, D, E, G, H, J



# BFS Traversal

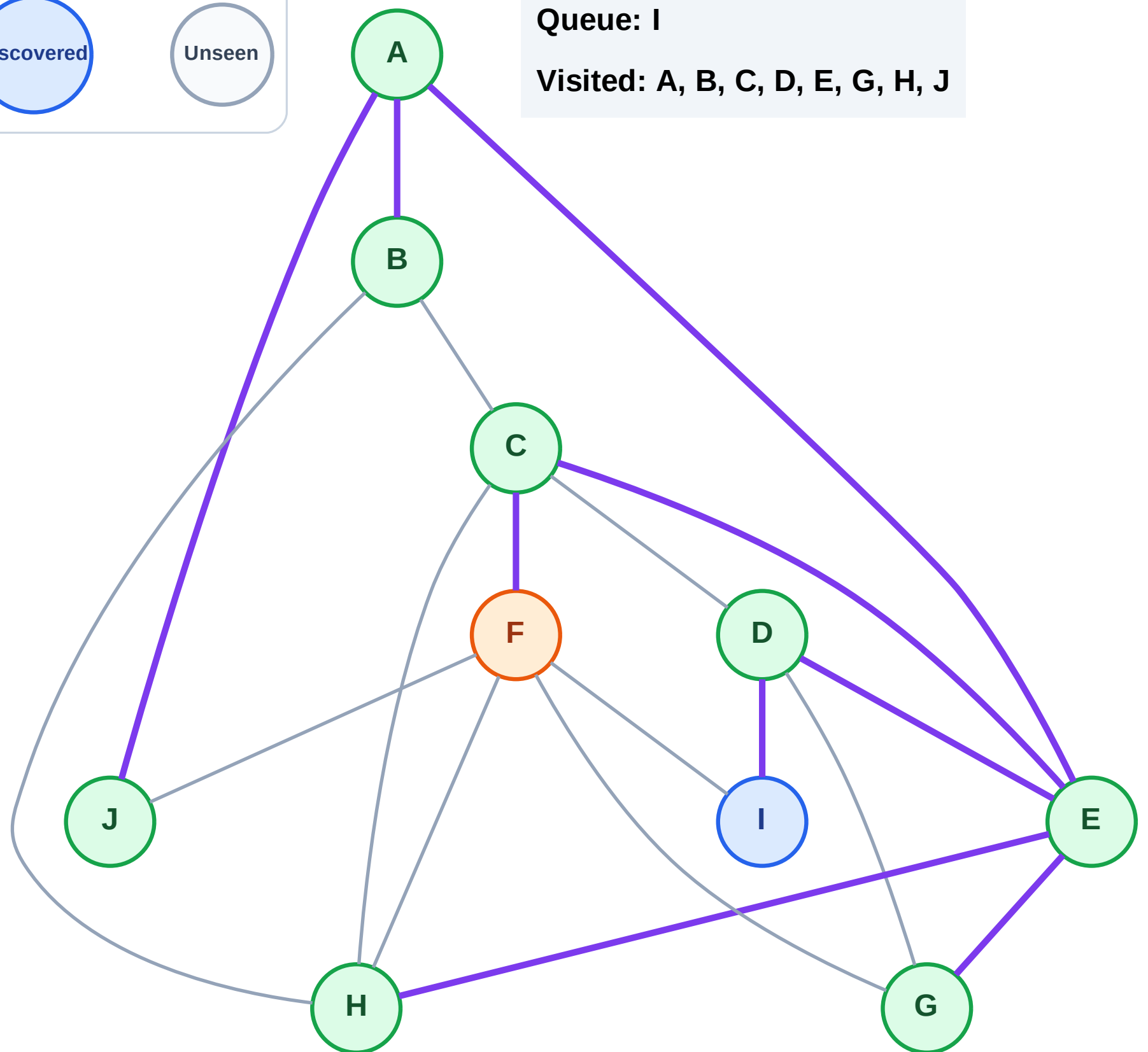
Step 18: Process node F

## Legend



Queue: I

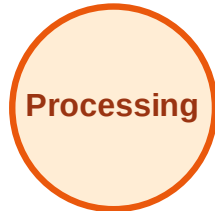
Visited: A, B, C, D, E, G, H, J



# BFS Traversal

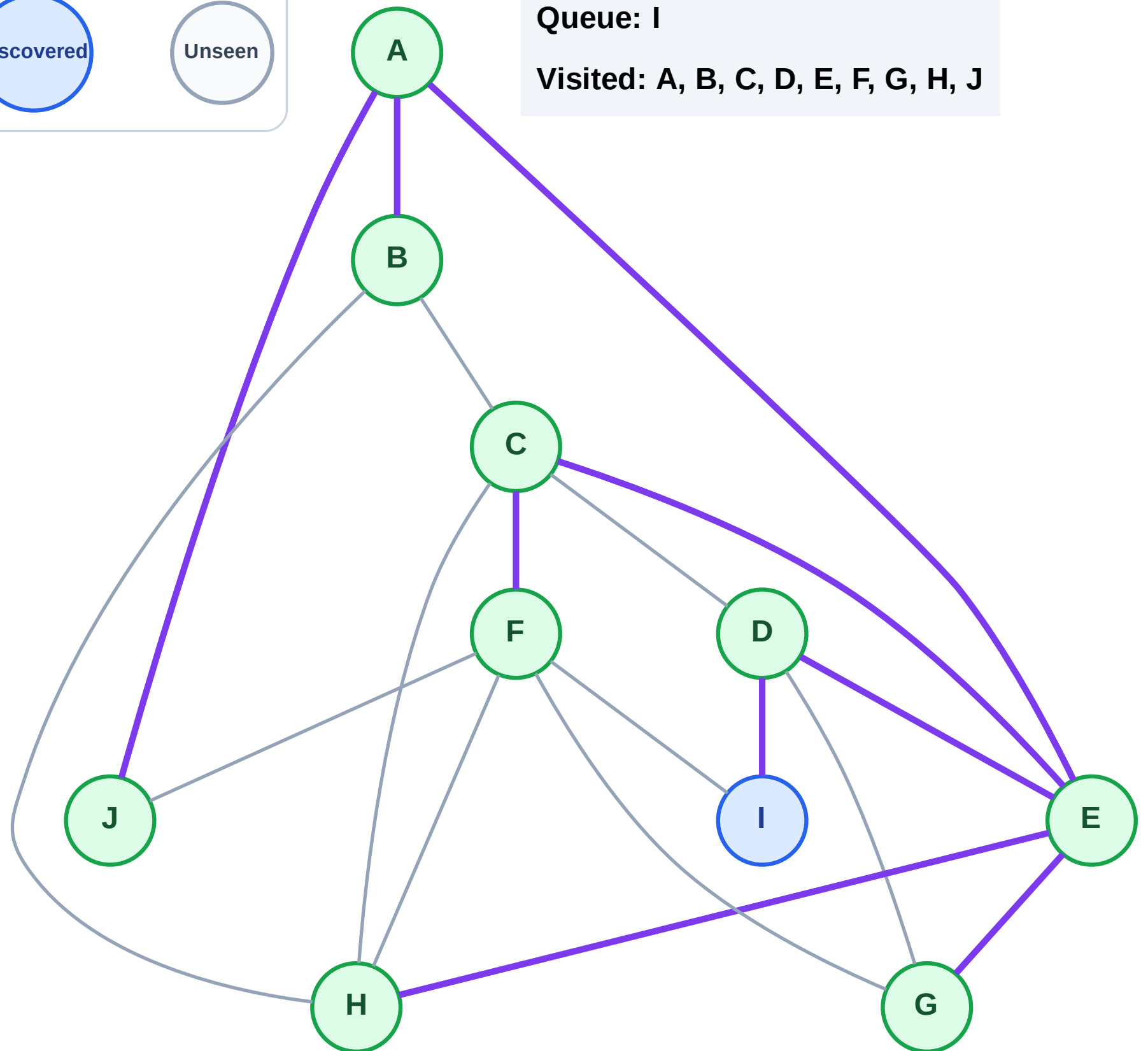
Step 19: No new neighbors from F

## Legend



Queue: I

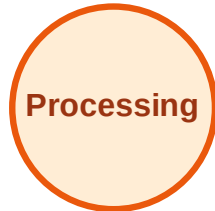
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

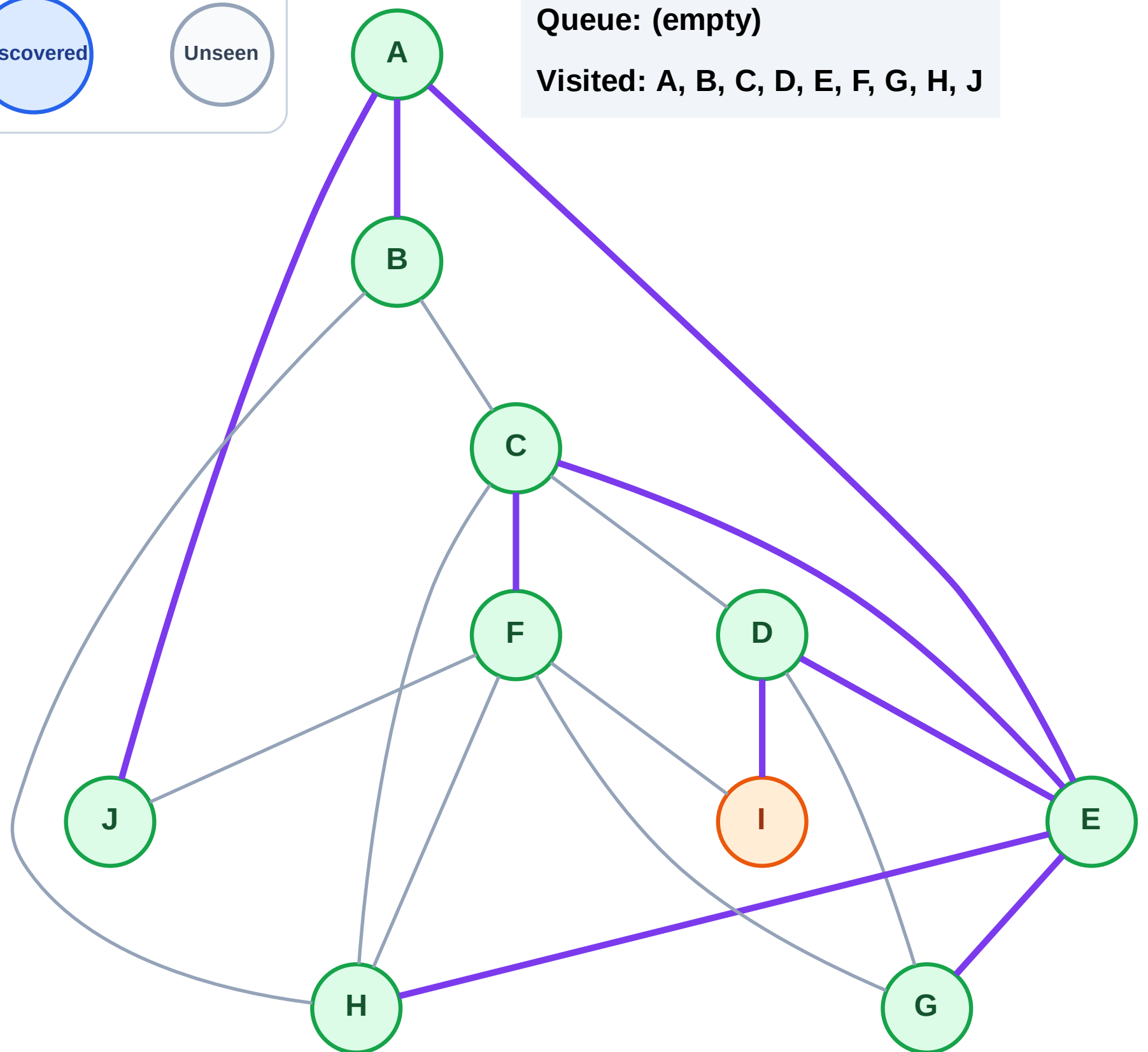
Step 20: Process node I

## Legend



Queue: (empty)

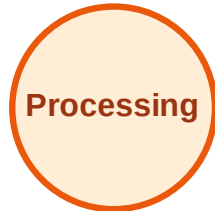
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

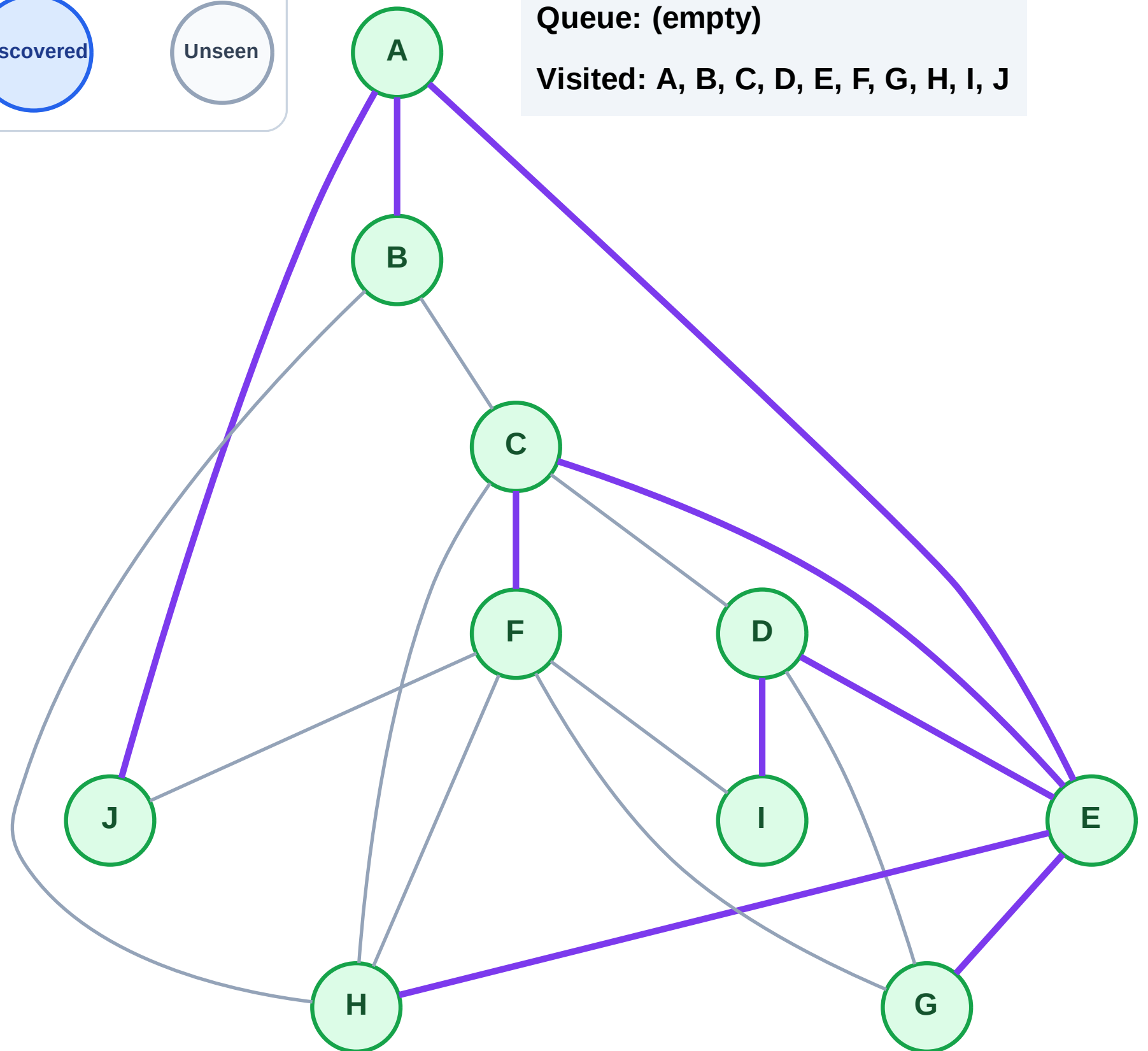
Step 21: No new neighbors from I

## Legend



Queue: (empty)

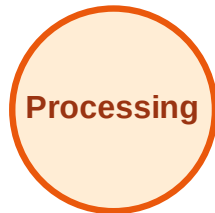
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

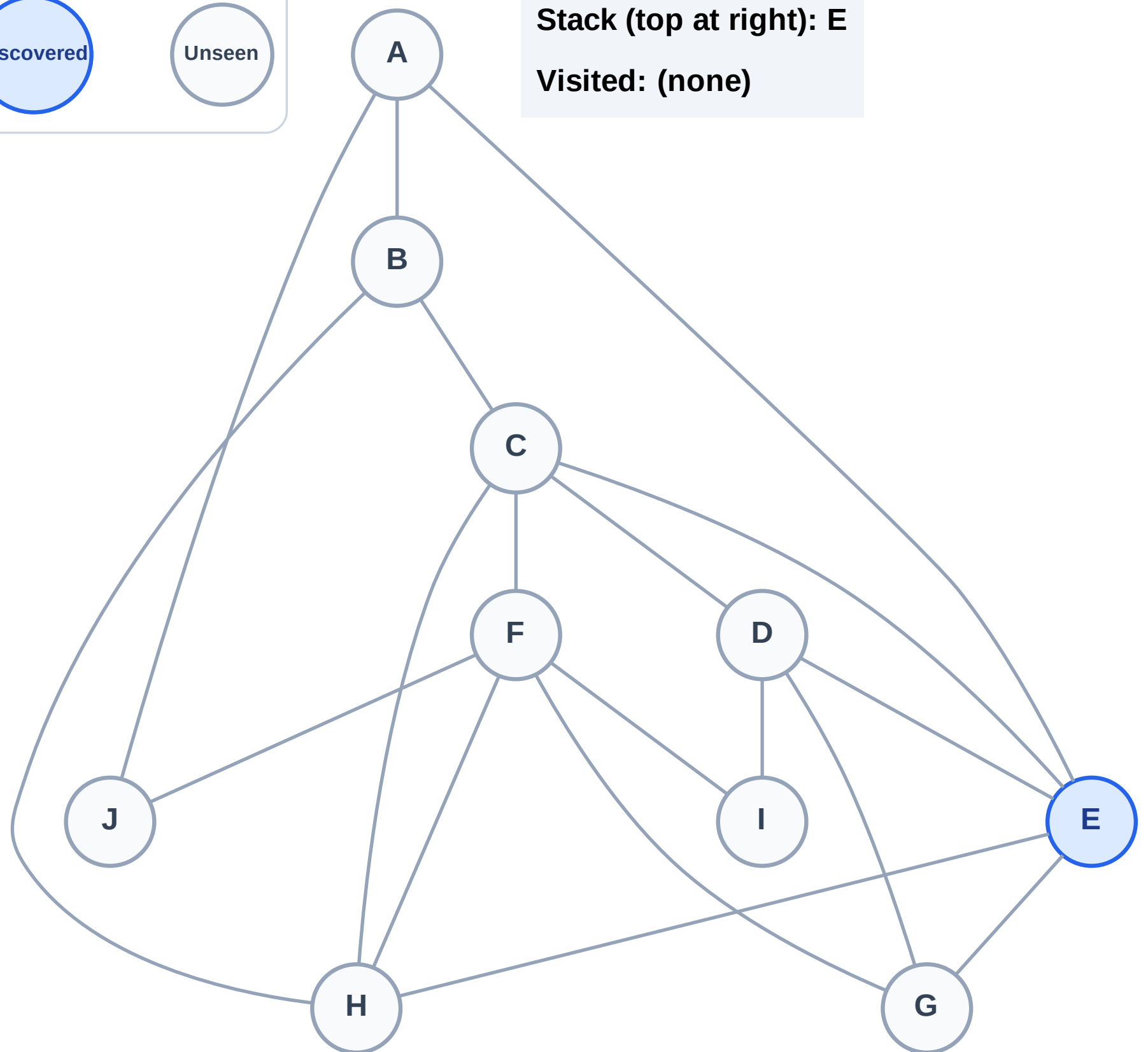
Step 1: Start at E

## Legend



Stack (top at right): E

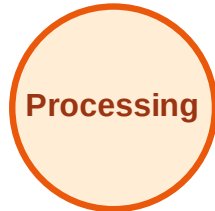
Visited: (none)



# DFS Traversal

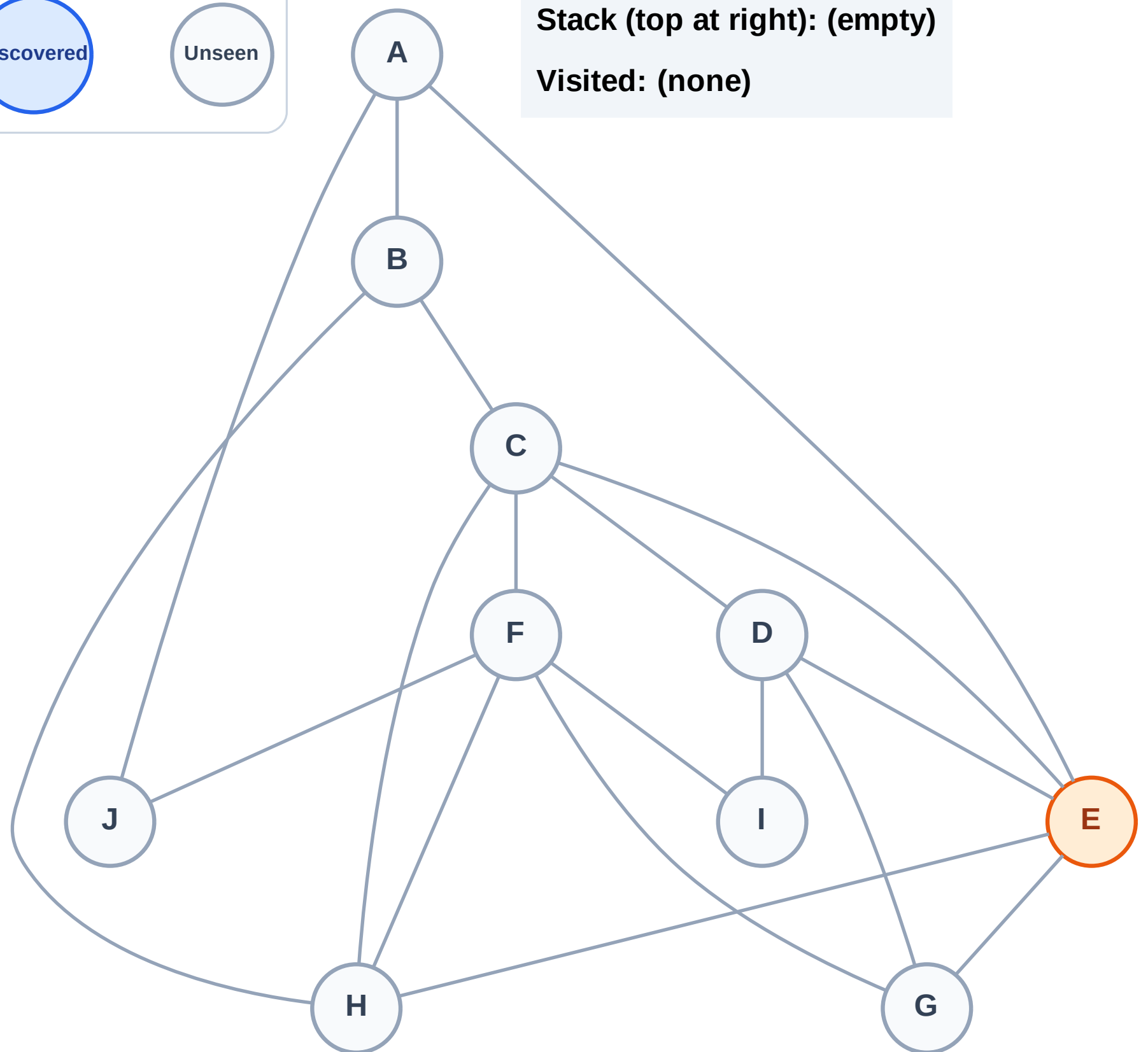
Step 2: Process node E

## Legend



Stack (top at right): (empty)

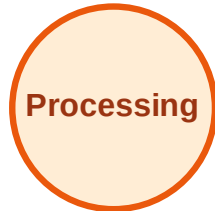
Visited: (none)



# DFS Traversal

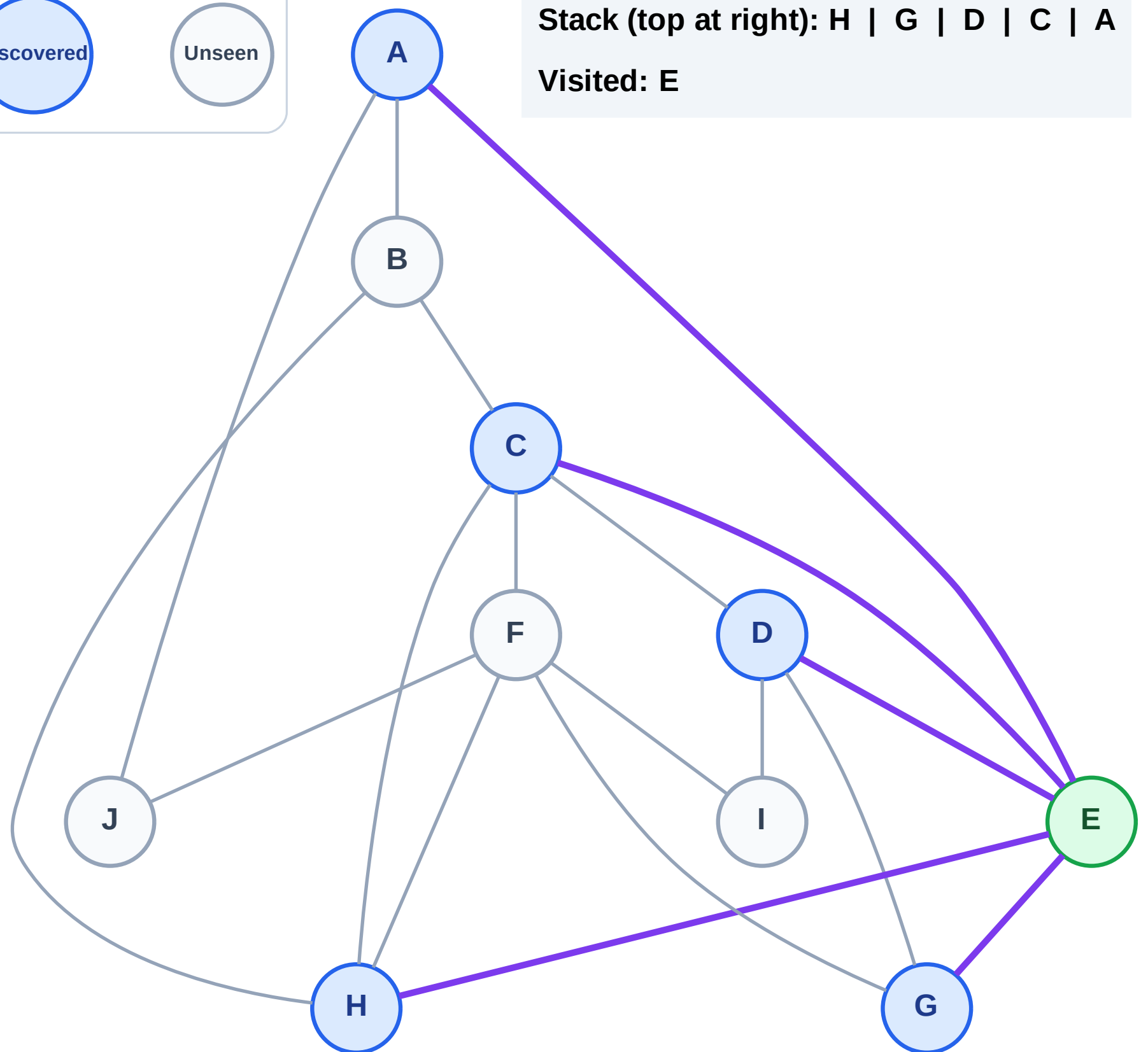
Step 3: Discover H, G, D, C, A from E

## Legend



Stack (top at right): H | G | D | C | A

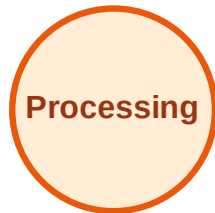
Visited: E



# DFS Traversal

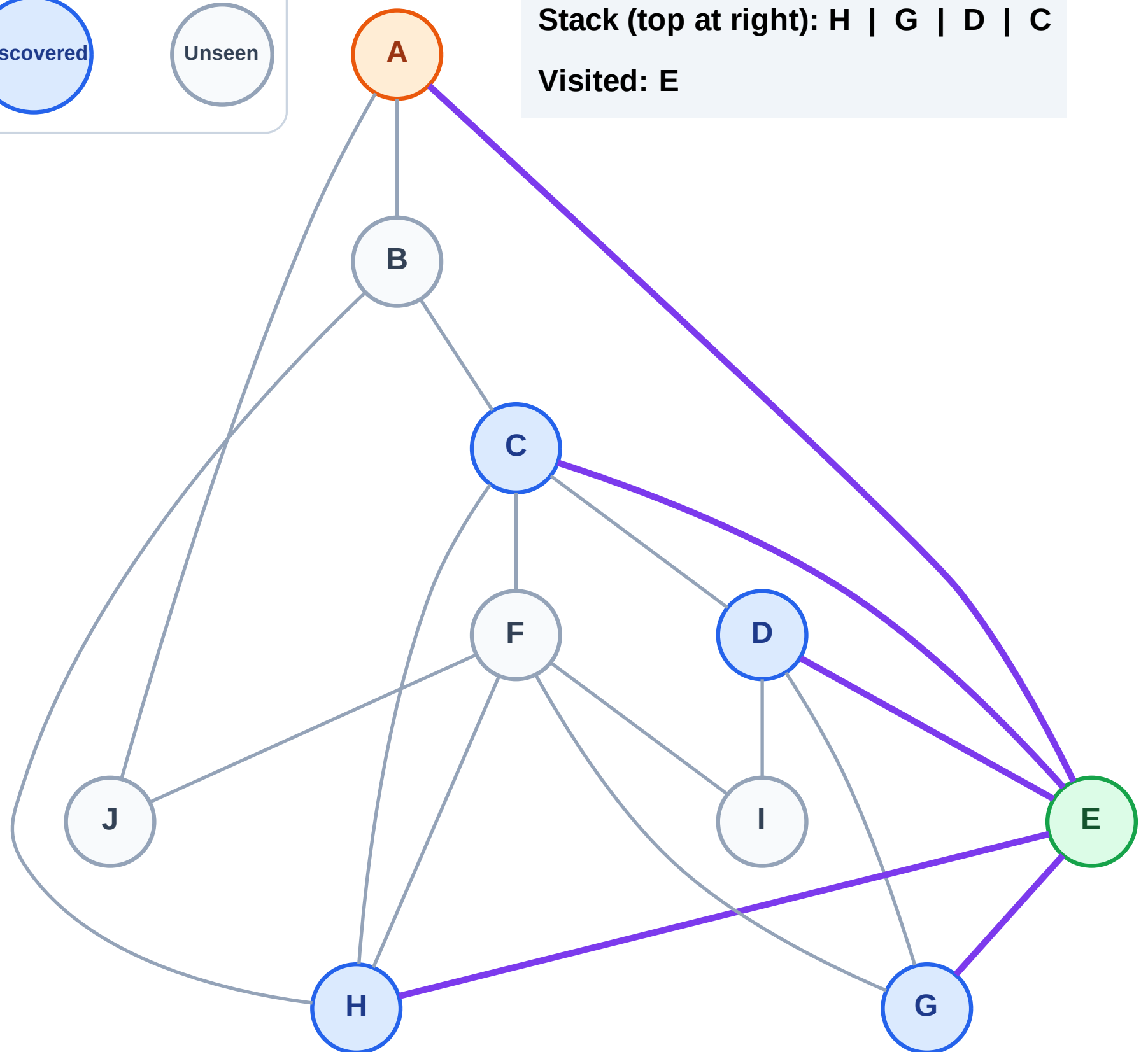
Step 4: Process node A

## Legend



Stack (top at right): H | G | D | C

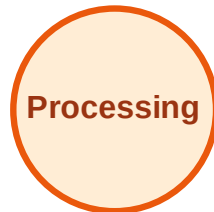
Visited: E



# DFS Traversal

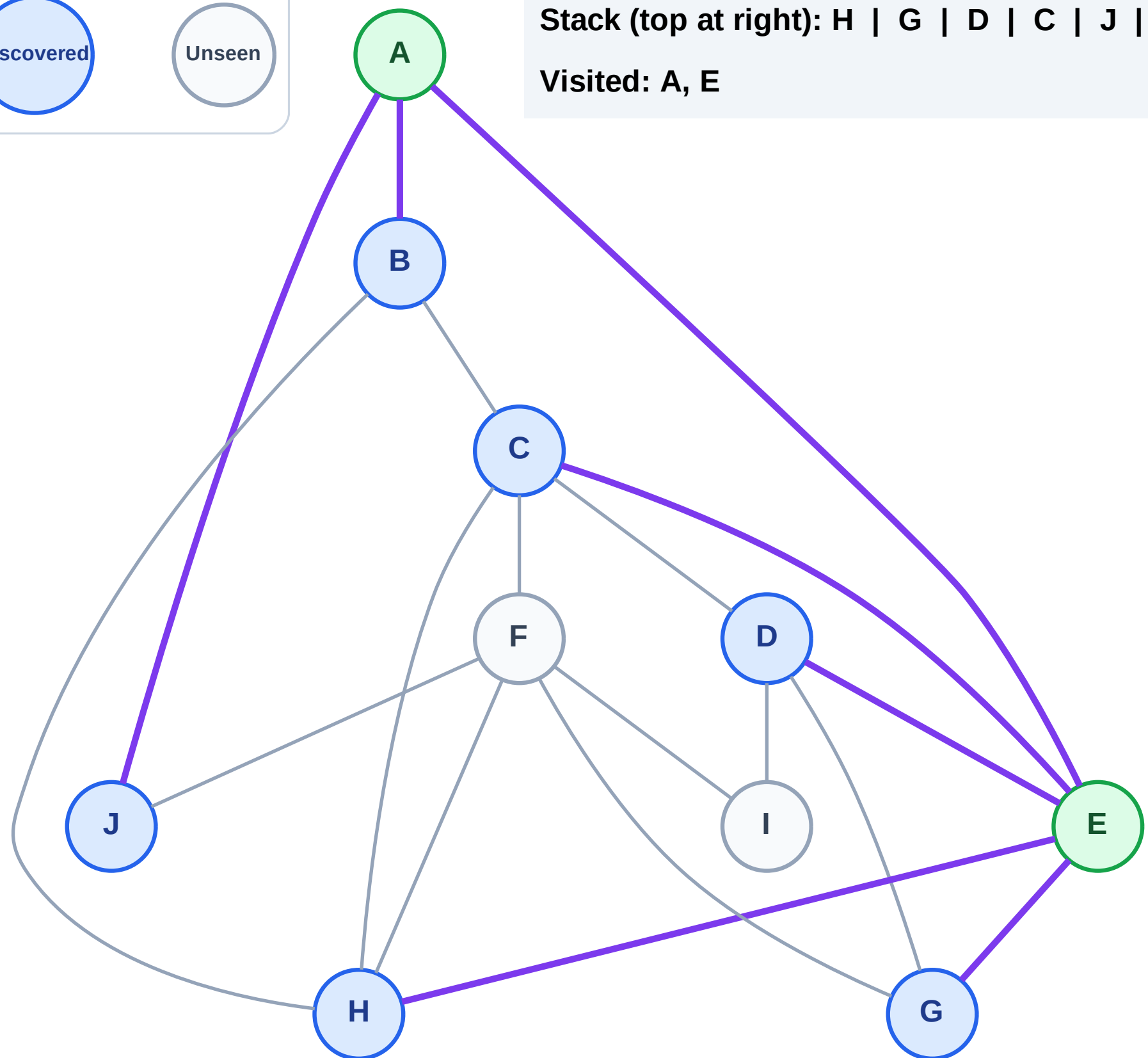
Step 5: Discover J, B from A

## Legend



Stack (top at right): H | G | D | C | J | B

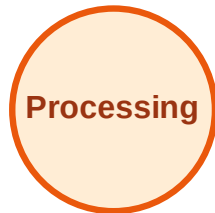
Visited: A, E



# DFS Traversal

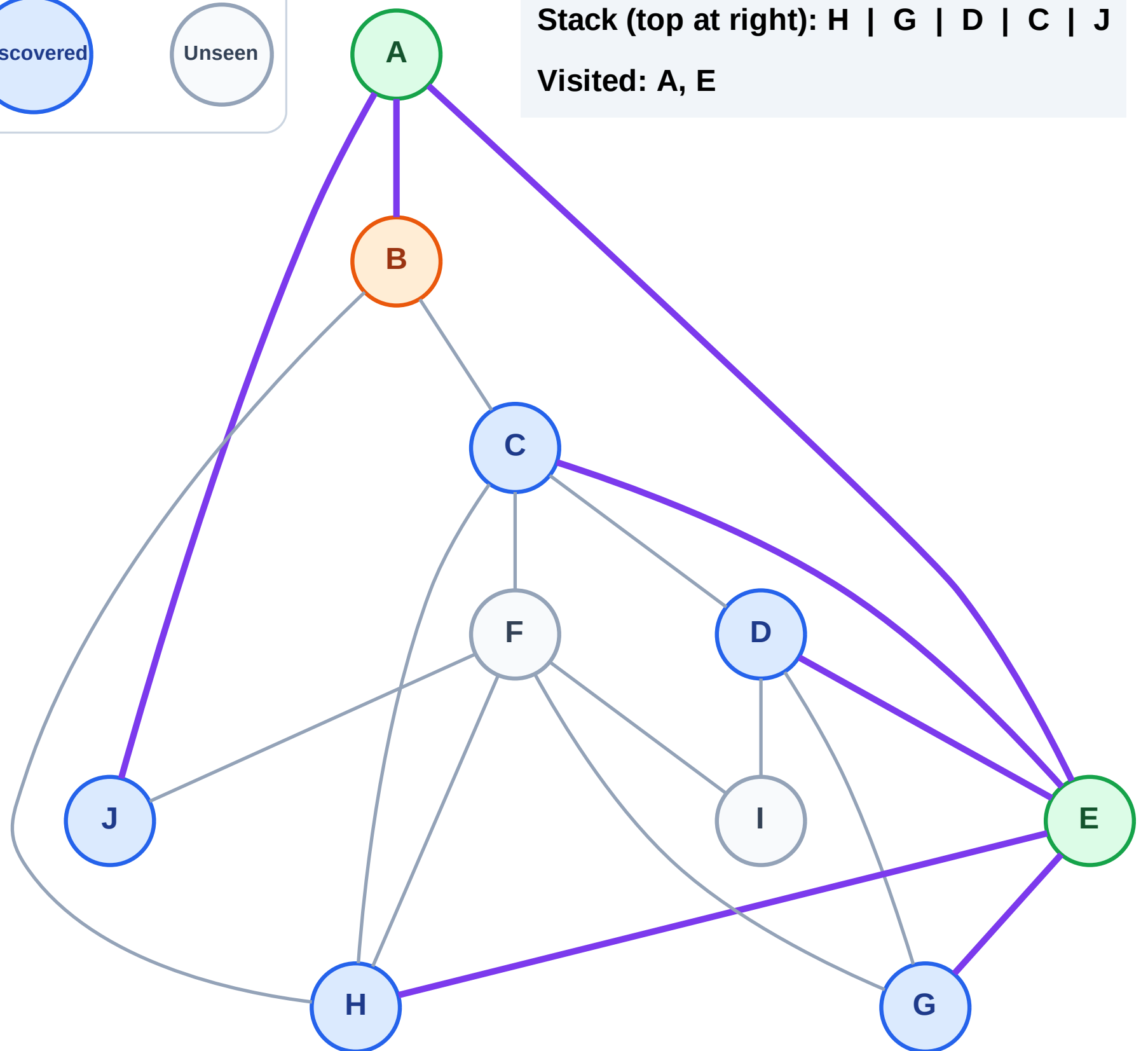
Step 6: Process node B

## Legend



Stack (top at right): H | G | D | C | J

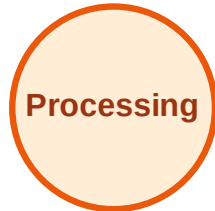
Visited: A, E



# DFS Traversal

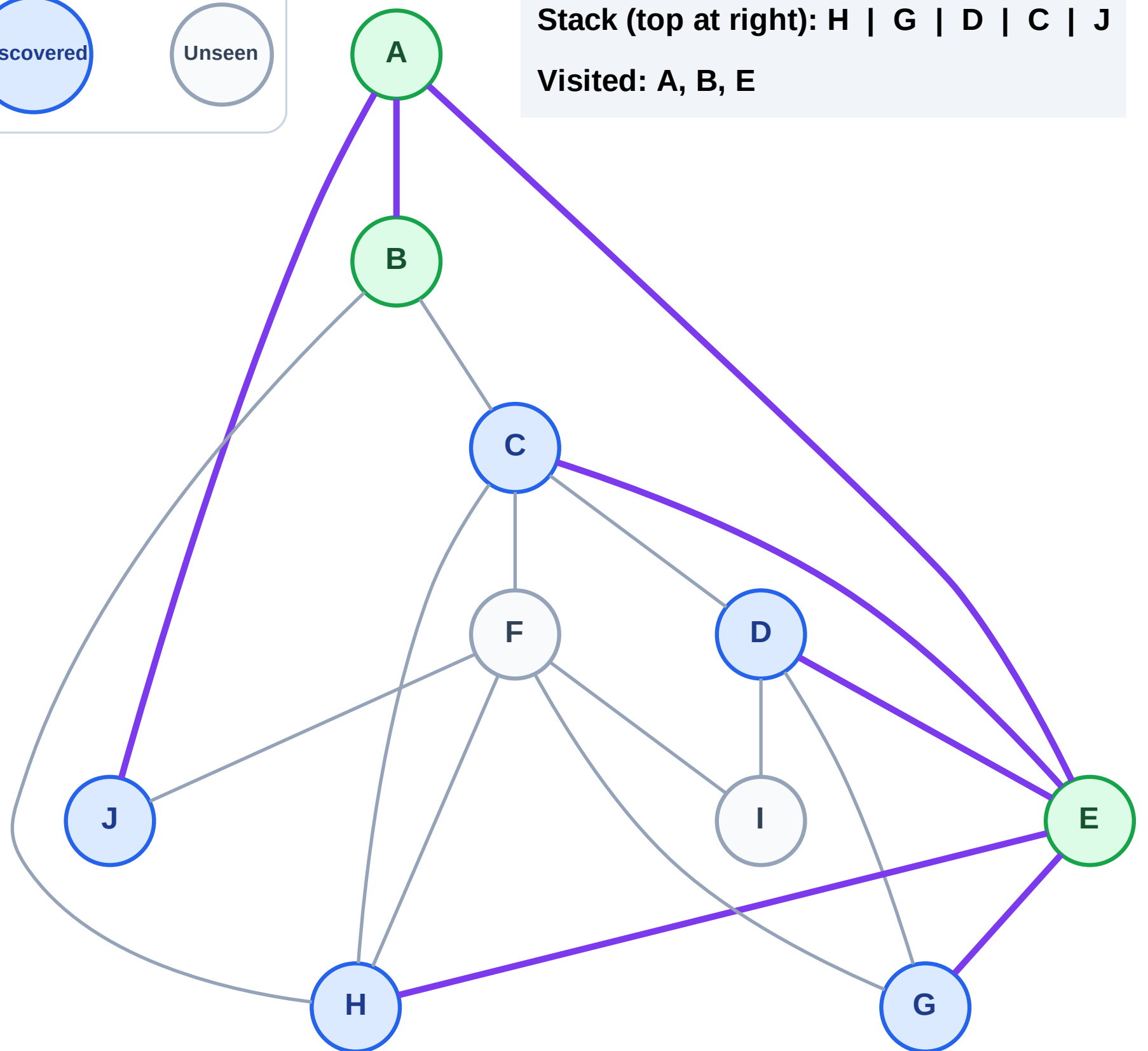
Step 7: No new neighbors from B

## Legend



Stack (top at right): H | G | D | C | J

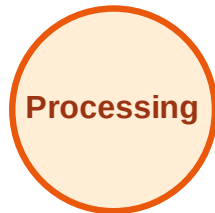
Visited: A, B, E



# DFS Traversal

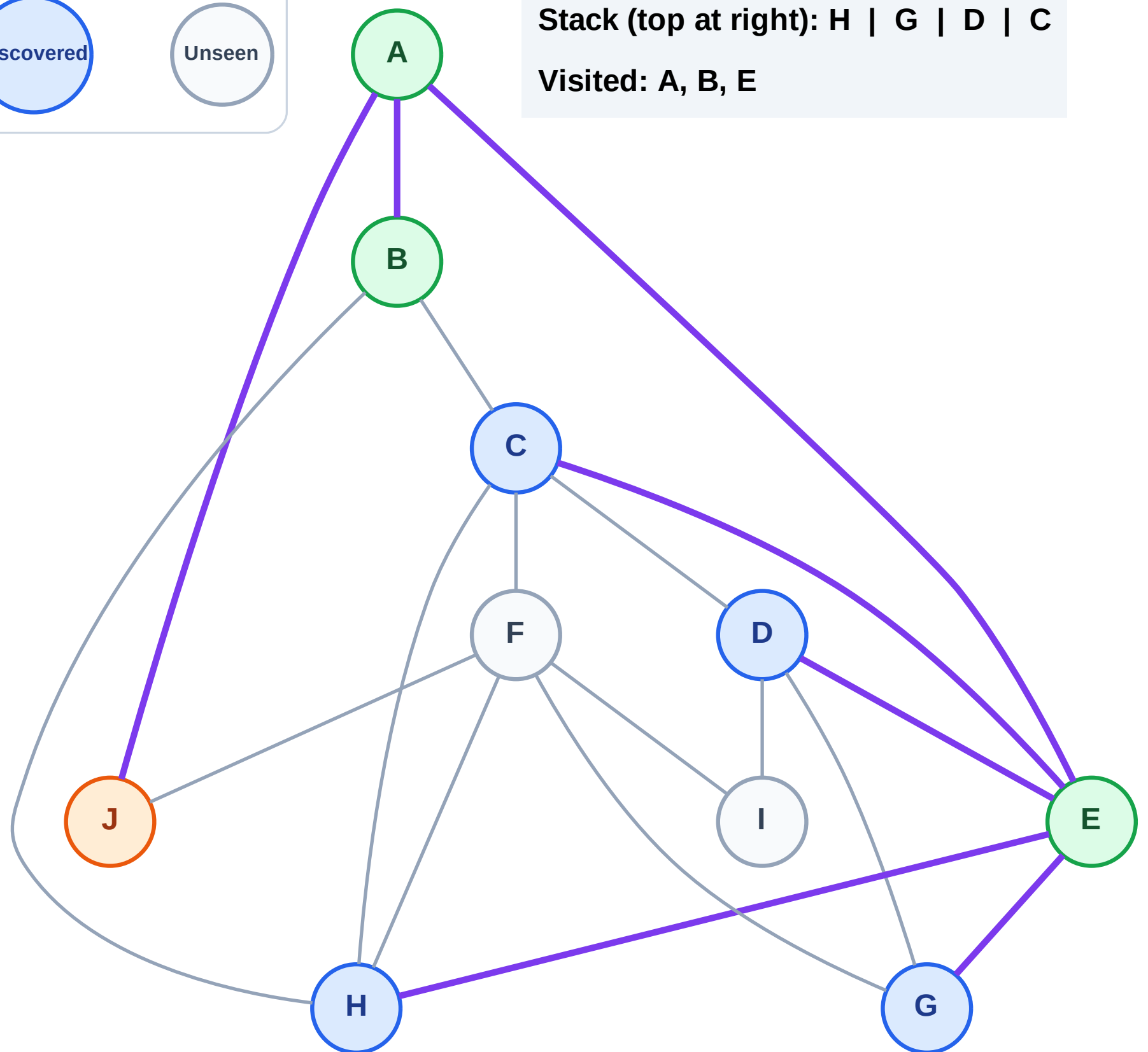
Step 8: Process node J

## Legend



Stack (top at right): H | G | D | C

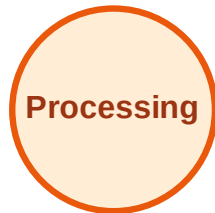
Visited: A, B, E



# DFS Traversal

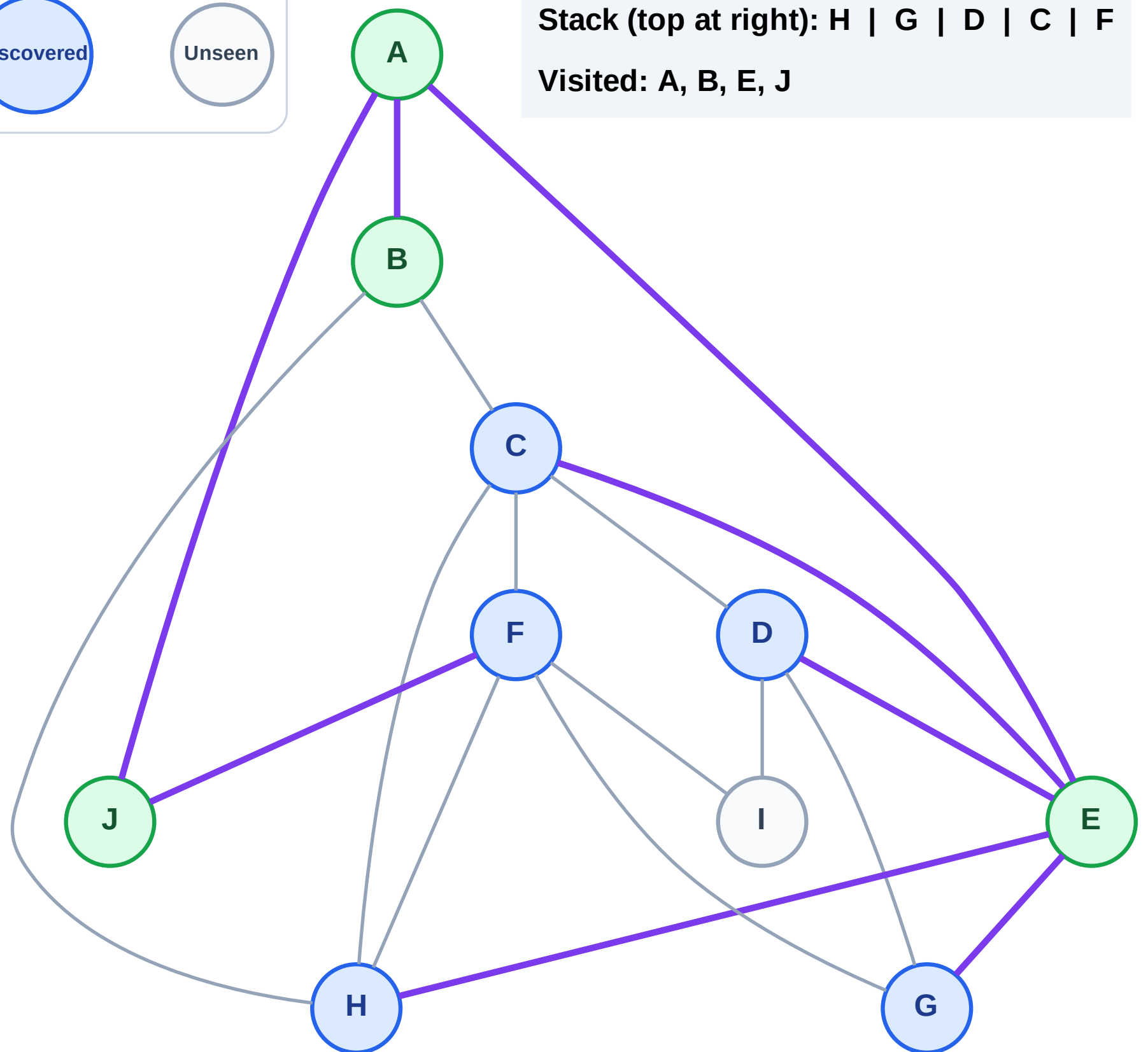
Step 9: Discover F from J

## Legend



Stack (top at right): H | G | D | C | F

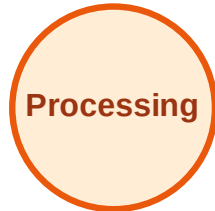
Visited: A, B, E, J



# DFS Traversal

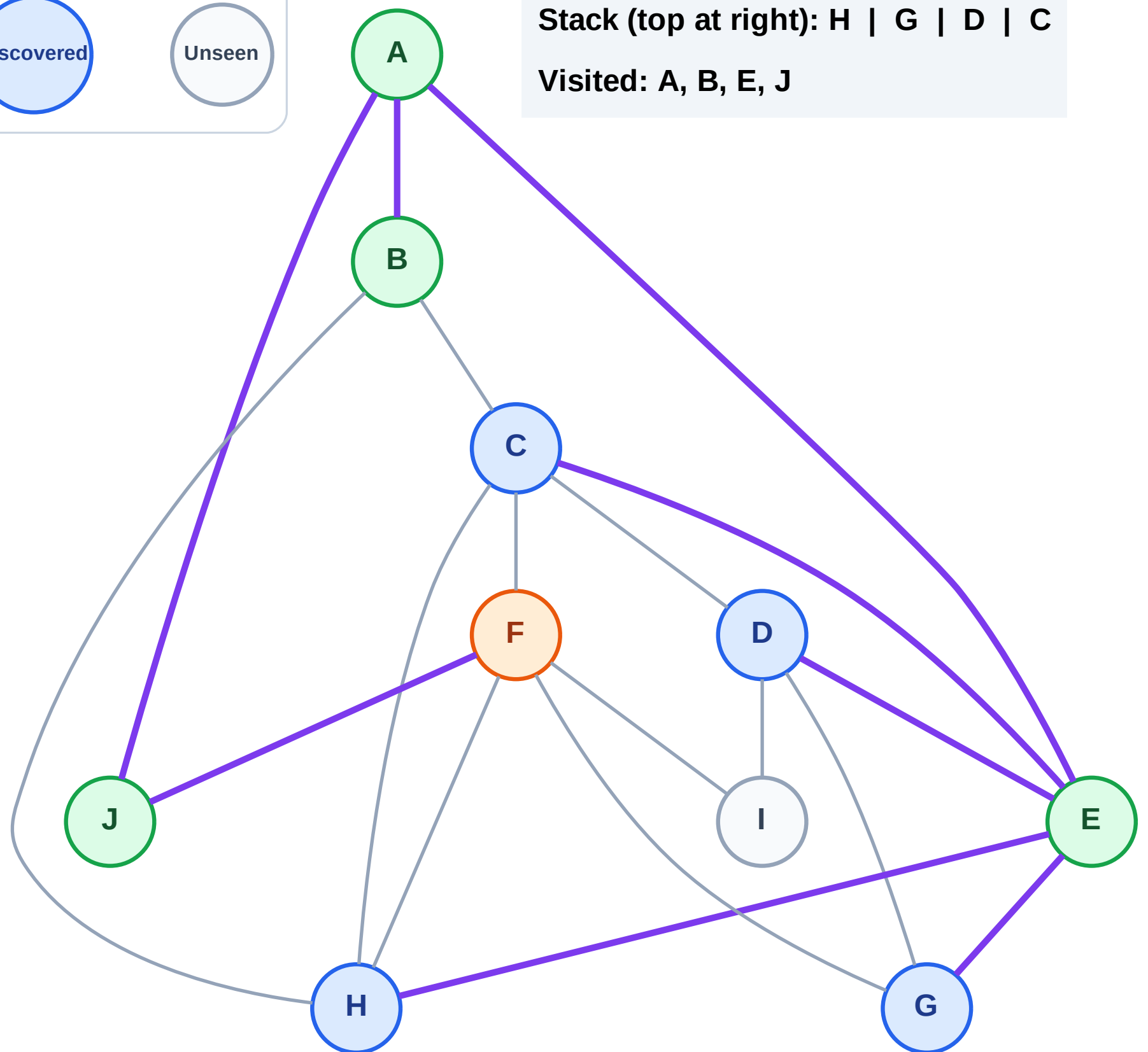
Step 10: Process node F

## Legend



Stack (top at right): H | G | D | C

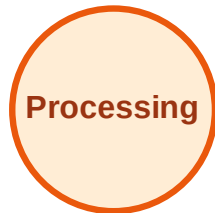
Visited: A, B, E, J



# DFS Traversal

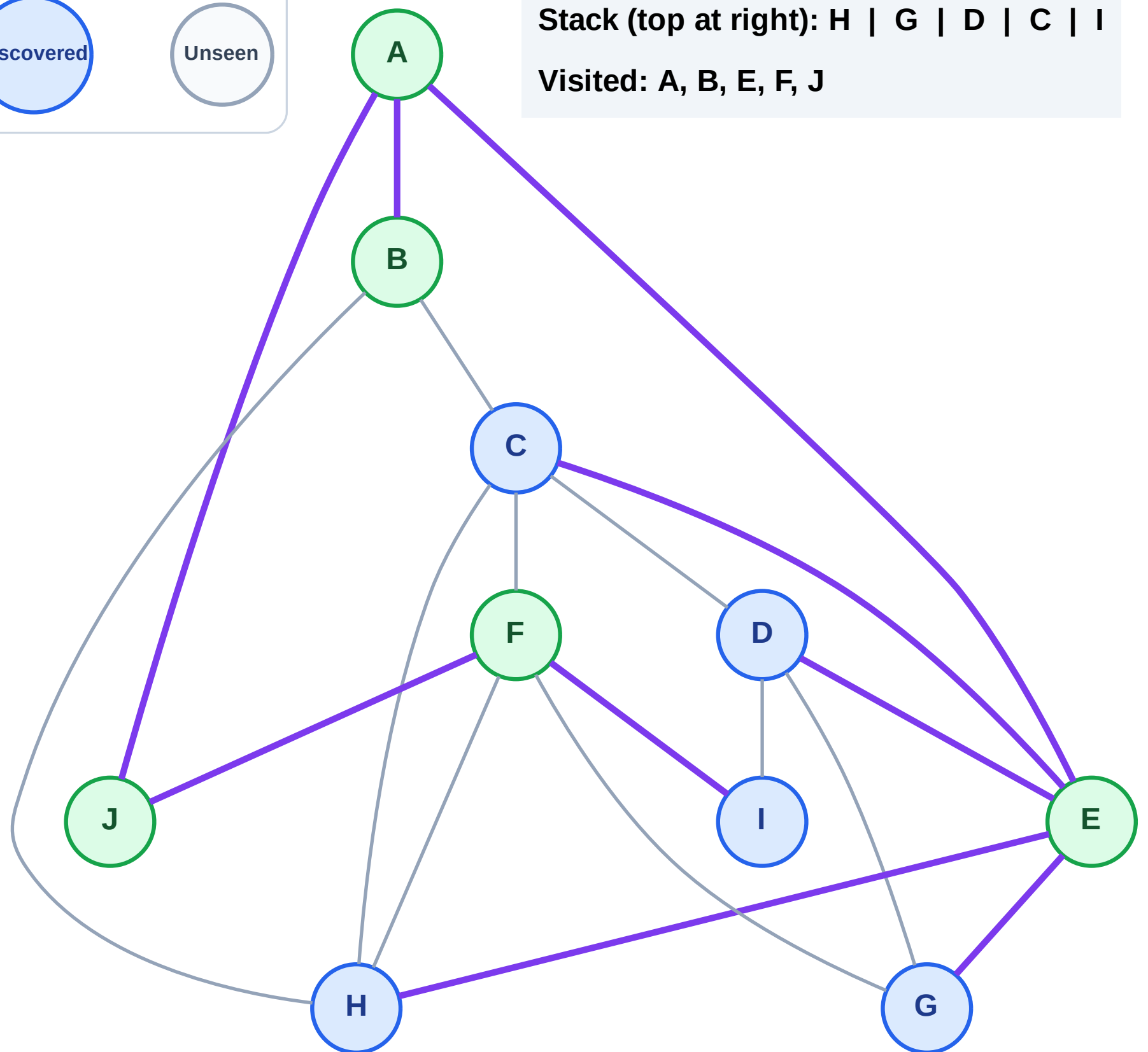
Step 11: Discover I from F

## Legend



Stack (top at right): H | G | D | C | I

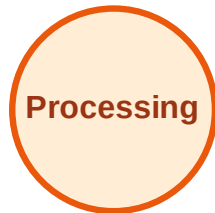
Visited: A, B, E, F, J



# DFS Traversal

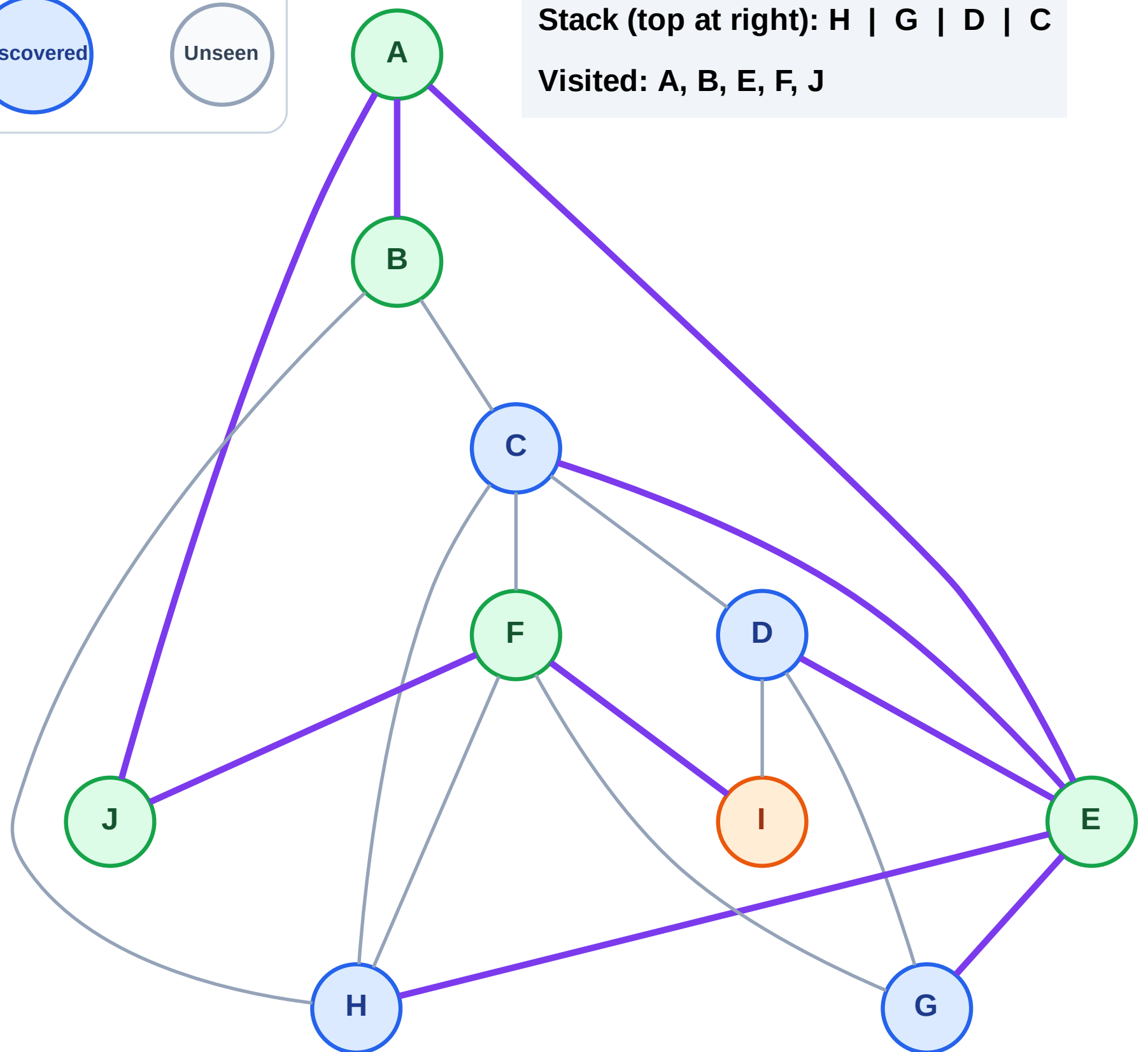
Step 12: Process node I

## Legend



Stack (top at right): H | G | D | C

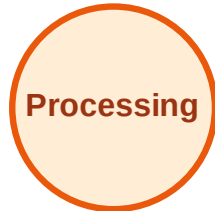
Visited: A, B, E, F, J



# DFS Traversal

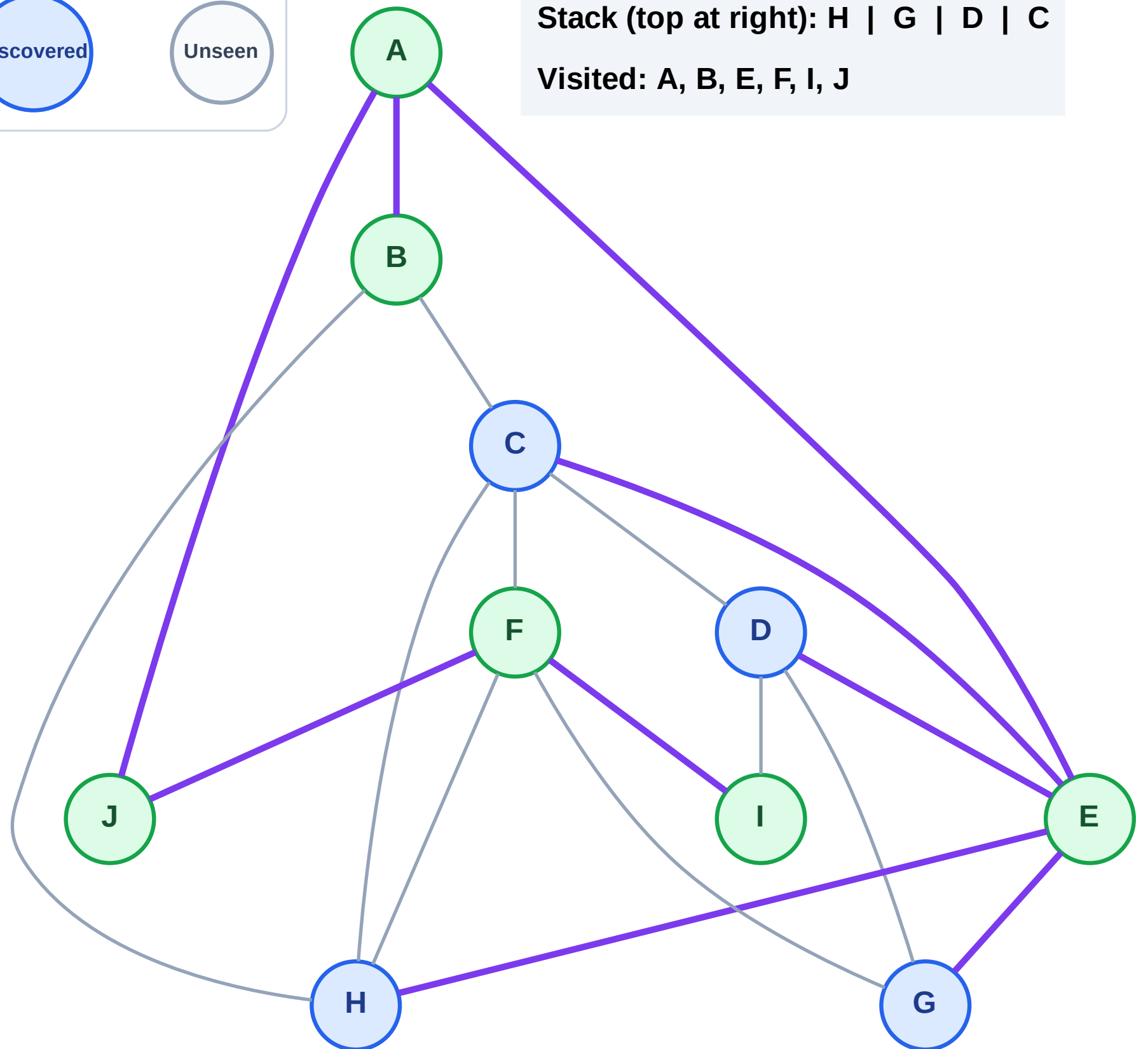
Step 13: No new neighbors from I

## Legend



Stack (top at right): H | G | D | C

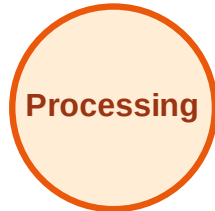
Visited: A, B, E, F, I, J



# DFS Traversal

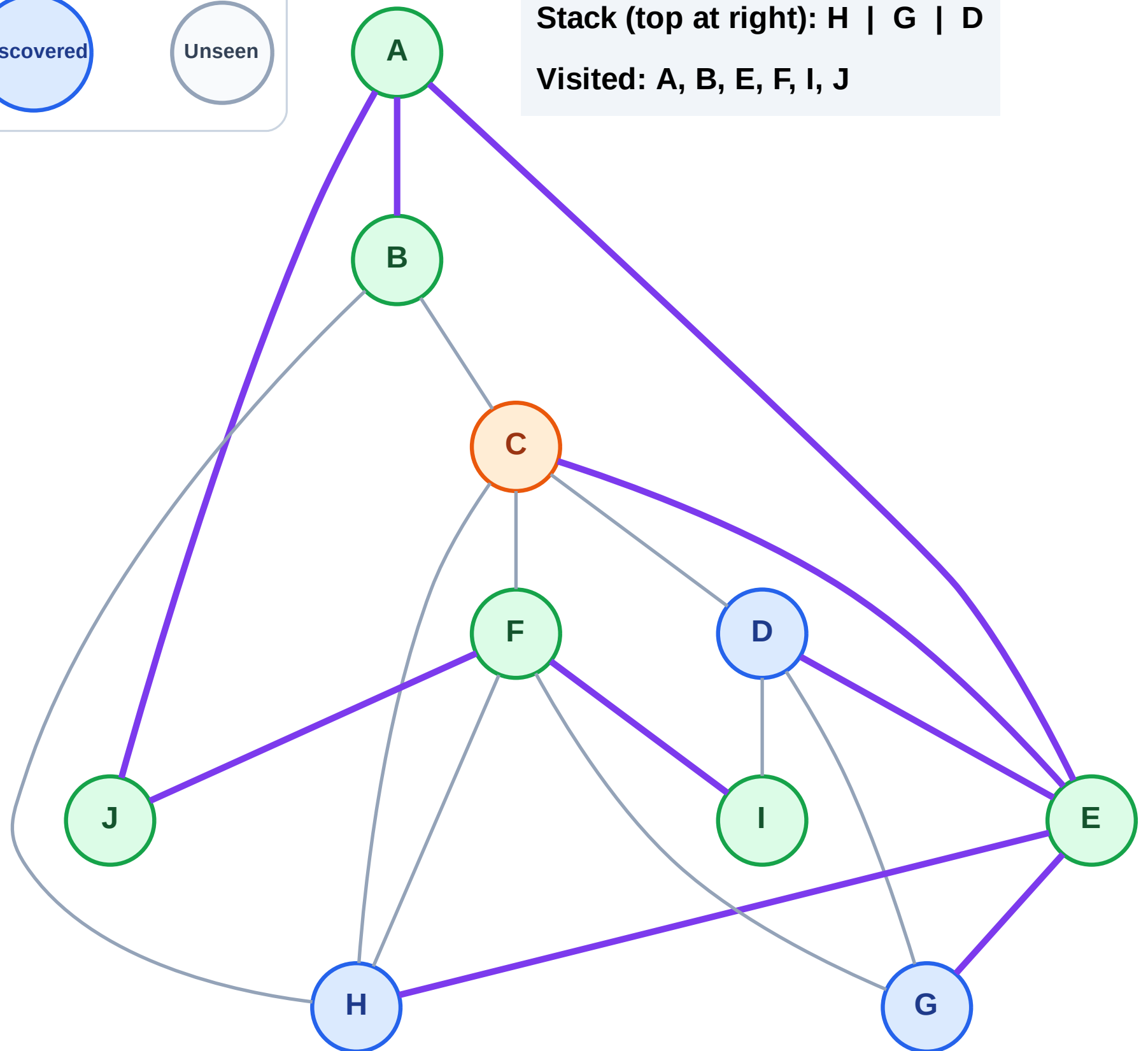
Step 14: Process node C

## Legend



Stack (top at right): H | G | D

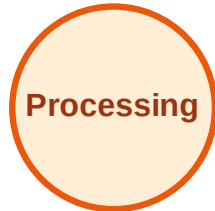
Visited: A, B, E, F, I, J



# DFS Traversal

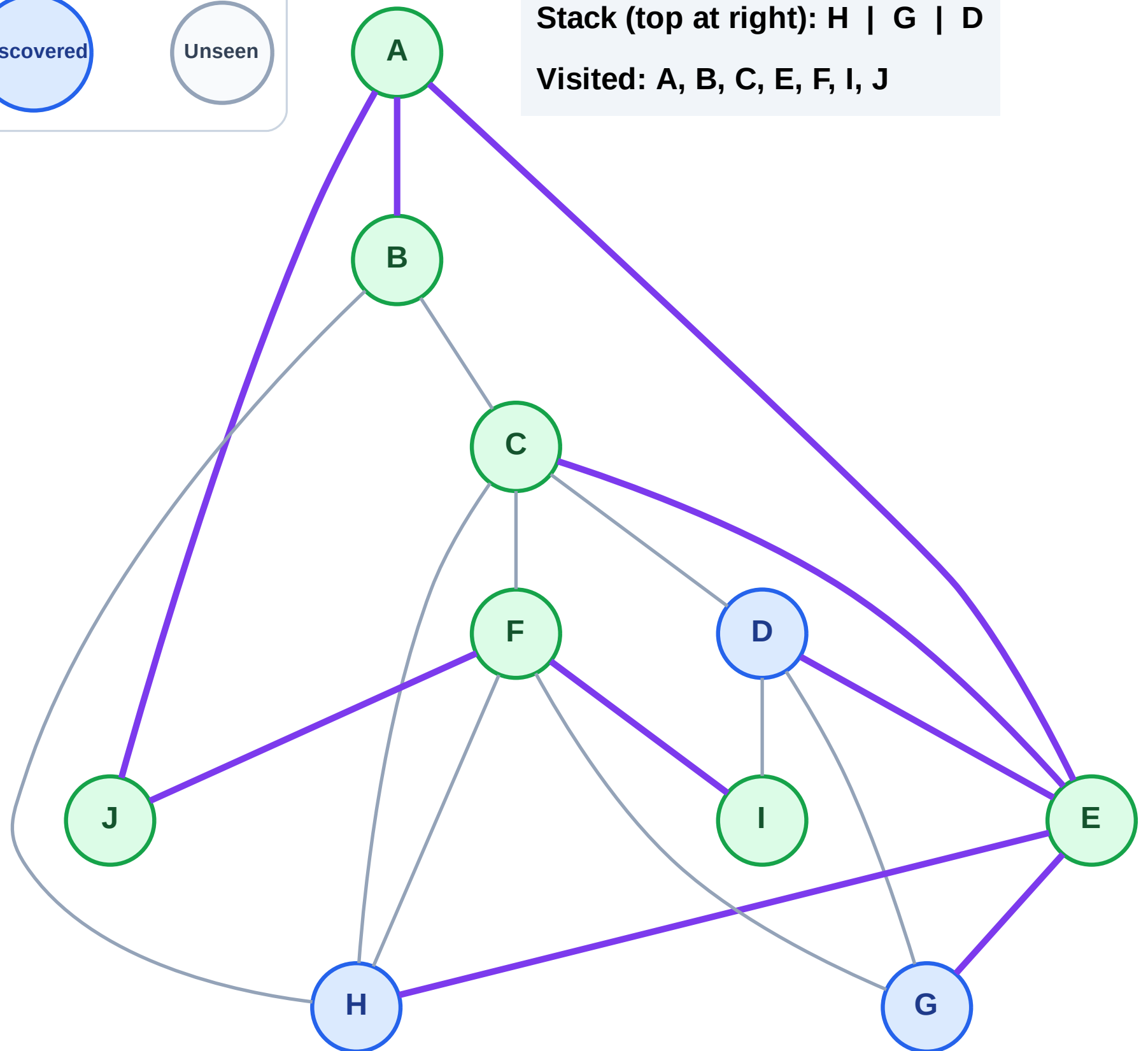
Step 15: No new neighbors from C

## Legend



Stack (top at right): H | G | D

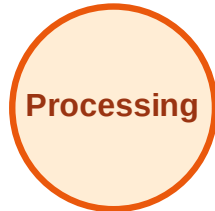
Visited: A, B, C, E, F, I, J



# DFS Traversal

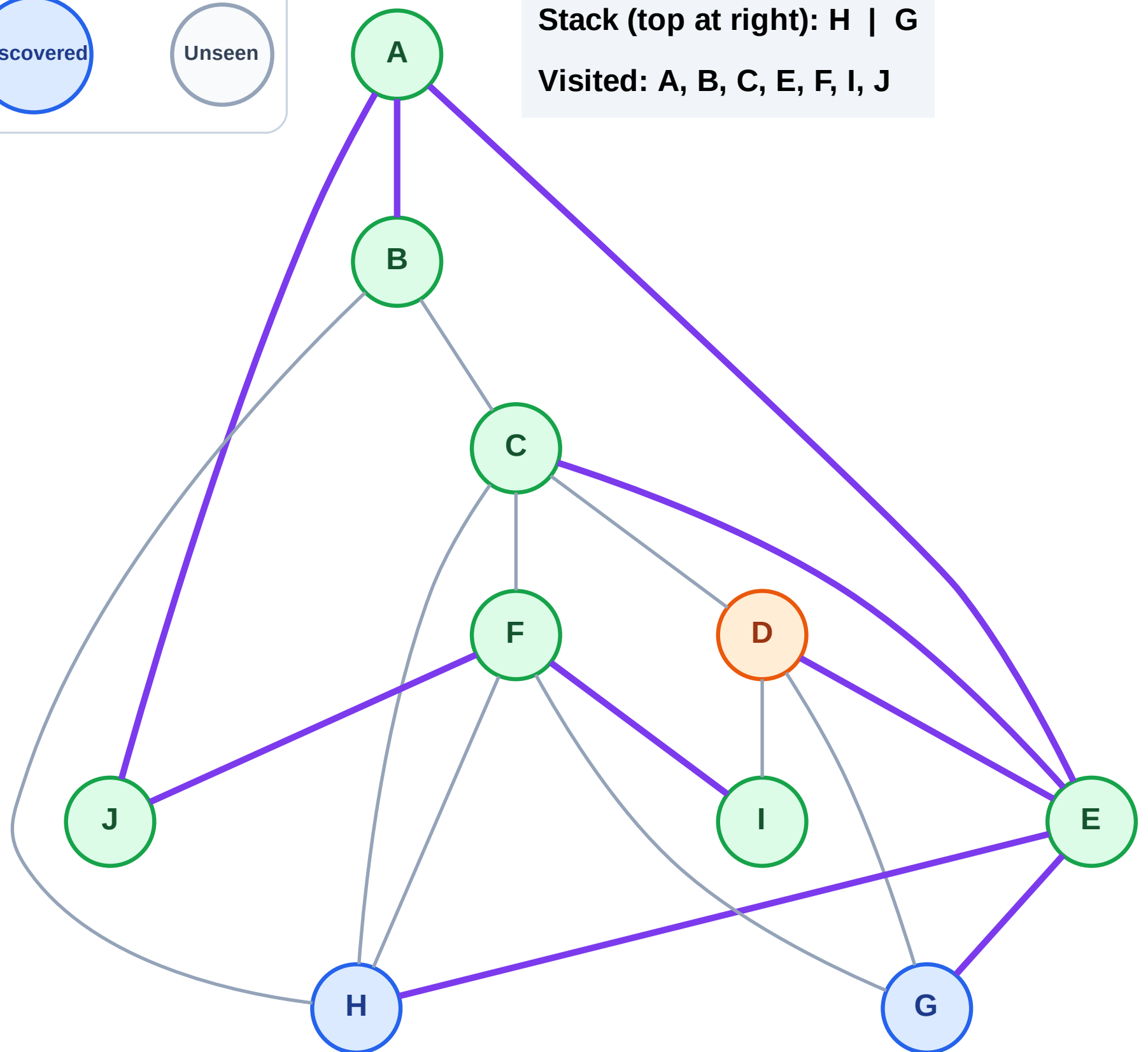
Step 16: Process node D

## Legend



Stack (top at right): H | G

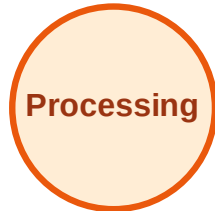
Visited: A, B, C, E, F, I, J



# DFS Traversal

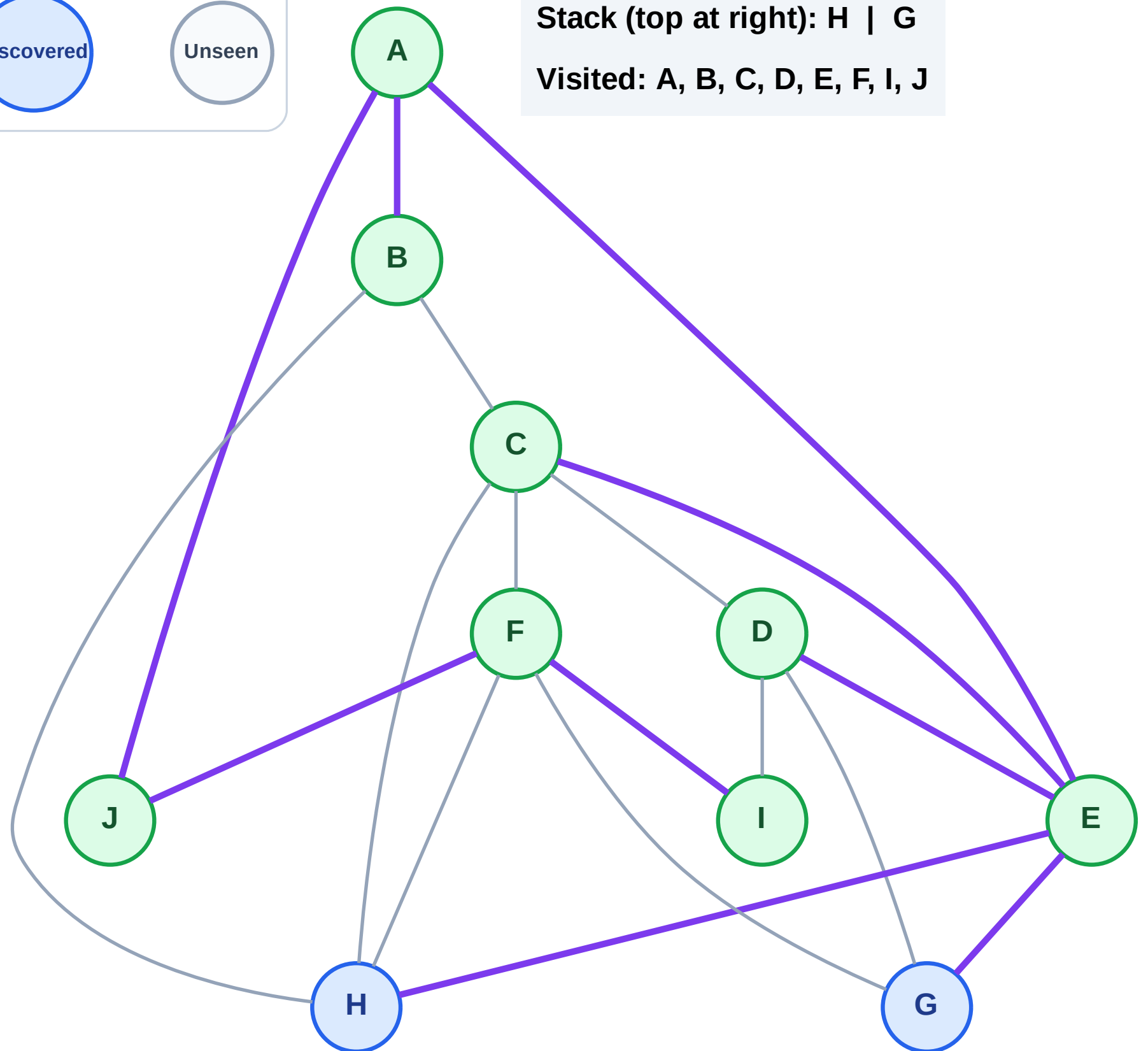
Step 17: No new neighbors from D

## Legend



Stack (top at right): H | G

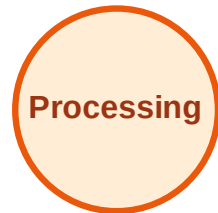
Visited: A, B, C, D, E, F, I, J



# DFS Traversal

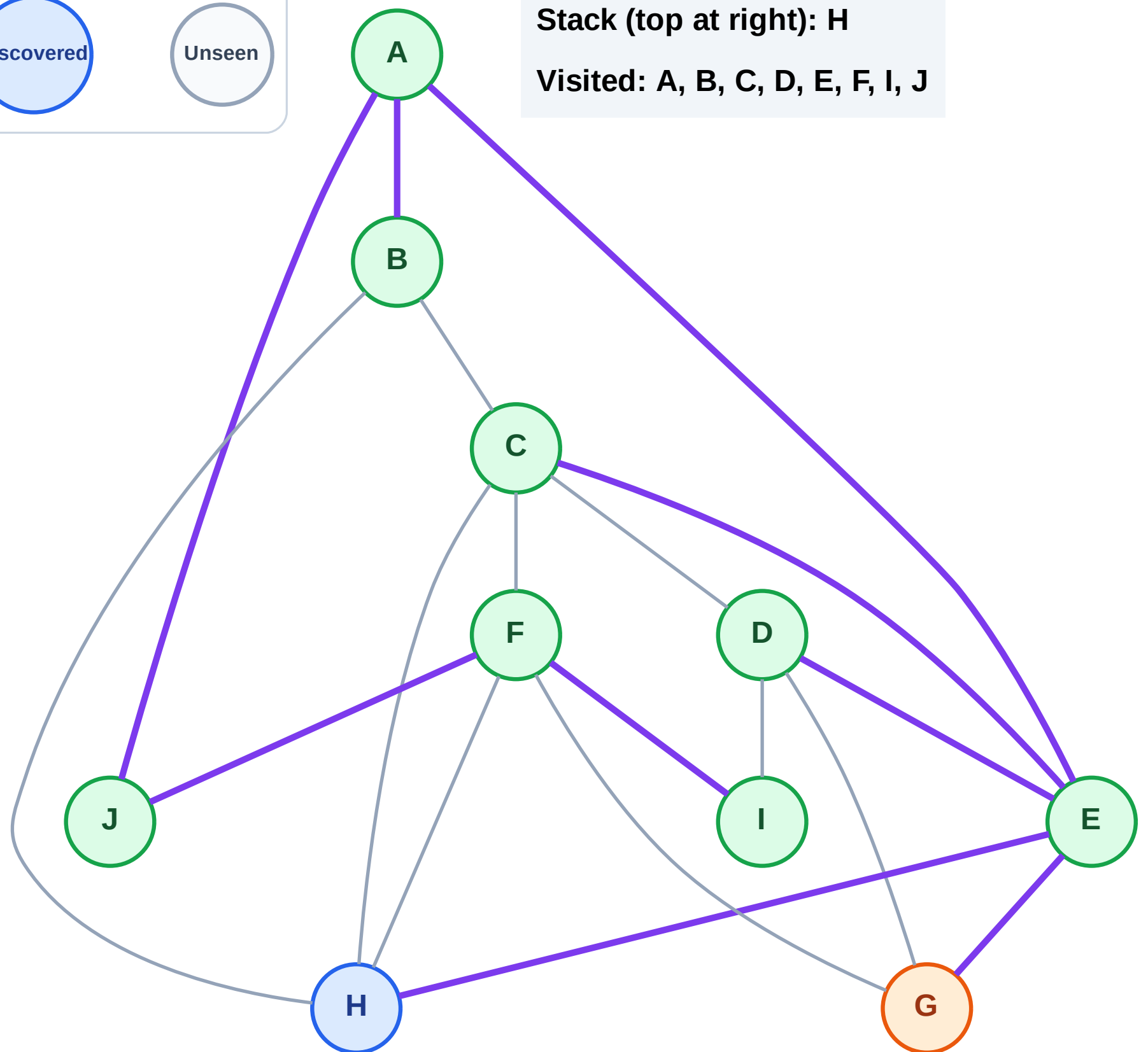
Step 18: Process node G

## Legend



Stack (top at right): H

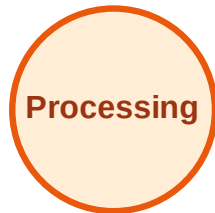
Visited: A, B, C, D, E, F, I, J



# DFS Traversal

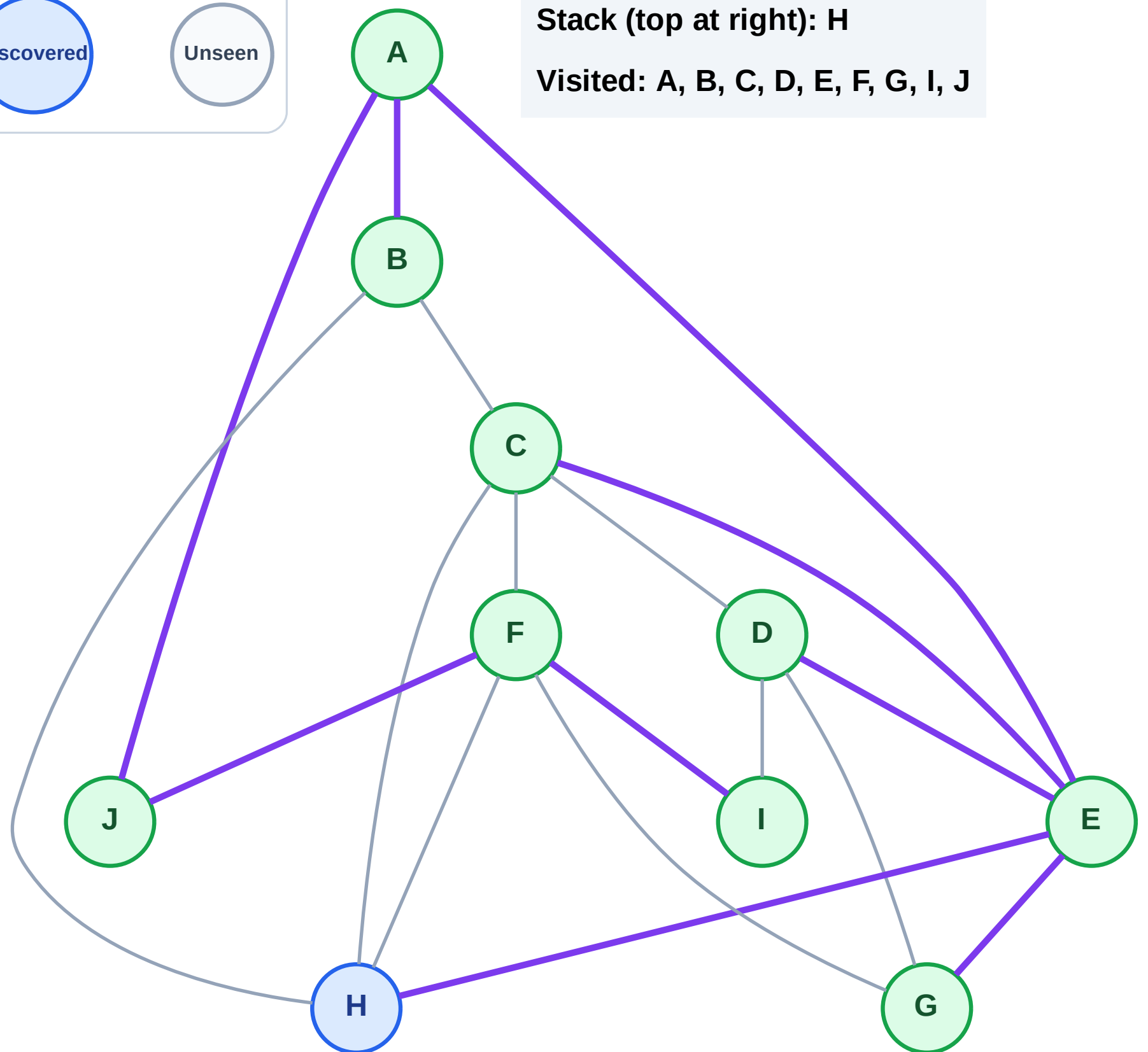
Step 19: No new neighbors from G

## Legend



Stack (top at right): H

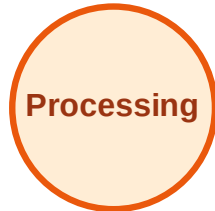
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

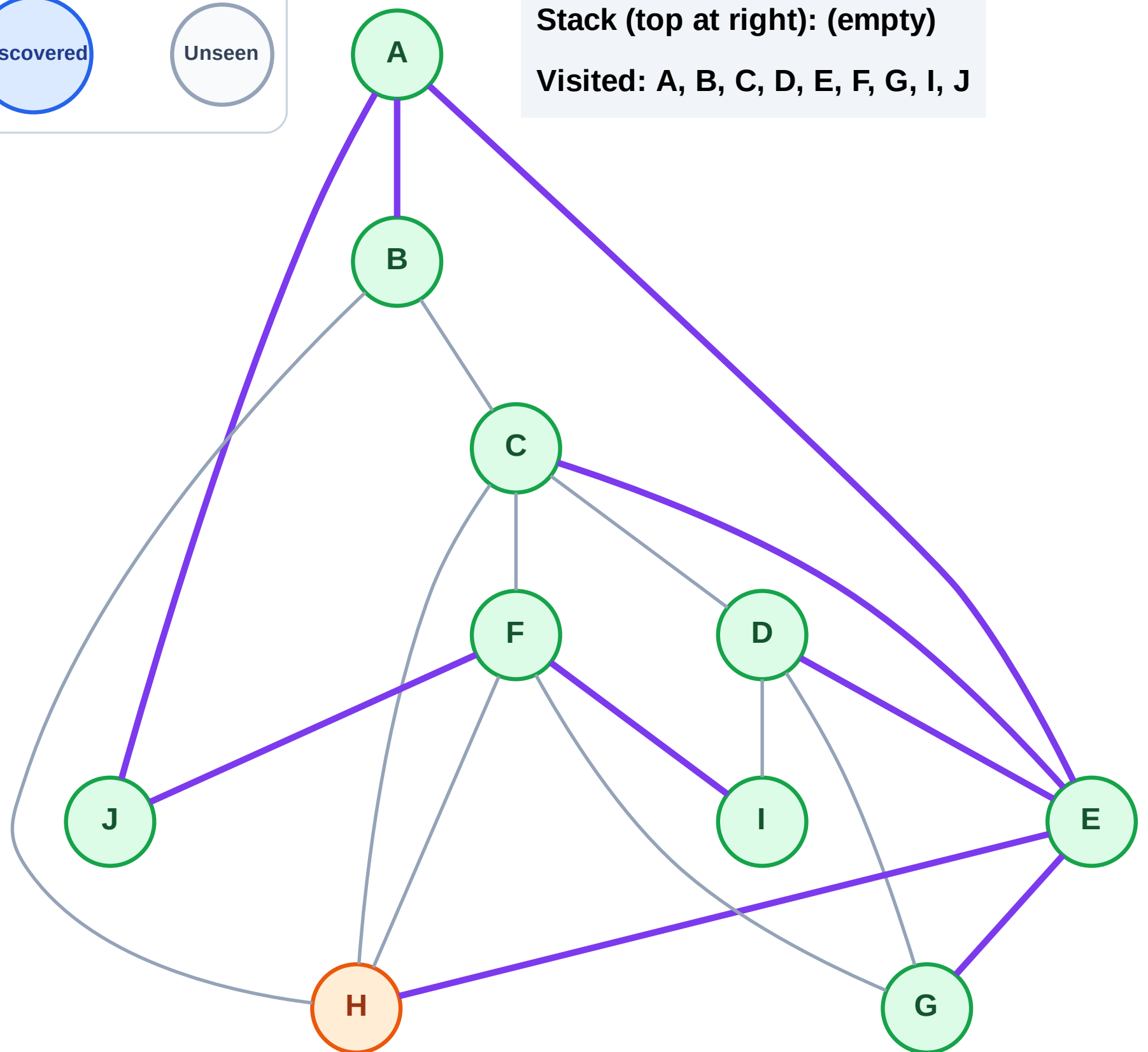
Step 20: Process node H

## Legend



Stack (top at right): (empty)

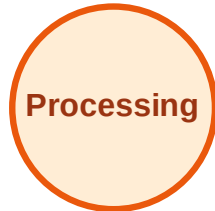
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

